

SECTION 4 – SITE ENVIRONMENT

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4-1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS

4-2-1 System Suite

Note

If these temperature and humidity specifications are not strictly adhered to, failures of the ACGD modules of the ACGD/PDU Cabinet may occur.

Use the specifications listed in Table 4-1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, make sure the magnet room temperature does not exceed 69.8°F (21°C) maximum.

TABLE 4-1
TEMPERATURE AND HUMIDITY SPECIFICATIONS

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Equipment Room at Inlet to Equipment	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)**
Magnet Room	59–69.8 (15–21)	5 (3)	30–60*	5	5 (3)
Operator's Control Room	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

4-2-2 Outdoor TSCC & TGWC Operating Environment

The Outdoor TSCC and Outdoor TGWC are designed to be located external to the building and operate in environments meeting the following specifications.

- Operating Ambient Temperature: –22°F (–30°C) to 122°F (50°C)
- Operating Humidity: 5–100%

4-3 AIR COOLING REQUIREMENTS

Table 4-2 contains the heat output of the equipment listed in the typical site location. These values do not include people, lights and non-MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Physical placement of the air conditioning equipment (compressor, etc.) is an important factor due to the homogeneous field requirements of the magnet. Therefore, it is important this equipment be located outside the 10 gauss line. Refer to Section 3, MAGNETIC FIELD CONSIDERATIONS, for plot of gauss lines.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal and Blower Box air intake for patient cooling.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner. Due to the large variation in heat loads, the compressors should be equipped with unloaders or hot gas bypass to prevent moisture stripping of the evaporator coils.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch high raised flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

4-3 AIR COOLING REQUIREMENTS (Continued)

TABLE 4-2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	MAGNET ROOM <i>See Note 1 below</i>		EQUIPMENT ROOM <i>See Note 2 below</i>		OPERATOR/CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly, Magnet Enclosure Equipment	8189	2400				
Patient Blower Box	1366	400				
Main Disconnect			900	264		
1.5T SRFD II Cabinet			18,423	5400		
ACGD/PDU Cabinet			34,120	10,000		
System Cabinet			6140	1800		
Twin Accessory Cabinet			2354	690		
TSCC Configurations for Magnet & Body Coil water cooling:						
• Outdoor Air Cooled Remote Control Panel (See Note 3 & 5)			34	10		
• Water Cooled (See Note 3 & 6)			12,000	3517		
• Indoor Air Cooled (See Note 3)			120,000	35,140		
TGWC Configurations for only Body Coil water cooling (requires site cooling for Shield Cooler Compressor):						
• Outdoor Air Cooled Remote Control Panel (See Note 4 & 5)			34	10		
• Water Cooled TGWC (See Note 4 & 6) RCP for water cooled TGWC			10,240	3000		
			34	10		
• Indoor Air Cooled (See Note 4)			61,000	17,880		
Shield/Cryo Cooler Compressor			Heat dissipation to air negligible			
Magnet Monitor			205	60		
Operator Workspace (See Note 7) with LCD Color Monitor					4095	1200
Magnet Monitor UPS* & Modem			450	132		
*MNS Cabinet			2730	800		
Note * Optional equipment. † Maximum heat output is defined for temperature and humidity as defined in Table 4-1. 1 Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room specifications as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contain a patient cooling fan. 2 FOR EQUIPMENT ROOM ONLY: The air cooling load averaged over a working day (~12 hours) is typically 1/2 of the maximum value. 3 The TSCC provides water cooling for the Shield/Cryo Cooler Compressor Cabinet (MS5) and the Gradient Coil. 4 The TGWC provides water cooling for the Gradient Coil only. When a TGWC is selected then the customer/site must still provide water cooling for the Shield/Cryo Cooler Compressor Cabinet. 5 The Outdoor Air Cooled TSCC and Outdoor Air Cooled TGWC are designed to be installed external to building and exhaust heat to outside air. A Remote Control Box is located in the Equipment Room. 6 The water cooled TSCC or TGWC require customer provided chilled water cooling in addition to the air cooling, refer to Section 4-4 Water Cooling Requirements for details. 7 Operator Workspace equipment includes the following: LCD Color Monitor, Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table						
(Continued)						

4-3 AIR COOLING REQUIREMENTS (Continued)

TABLE 4-3
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	MAGNET ROOM <i>See Note 1 below</i>		EQUIPMENT ROOM <i>See Note 2 below</i>		OPERATOR/CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
*Advantage Workstation Option					See Note 8	
*BrainWave Audio Console					136	40
*BrainWave Cabinet			5118	1500		
*BrainWave Cart	1638	480				
Note * Optional equipment. 8 Refer to Direction 2261309-100 <i>Advantage Workstation Ultra60 Pre-Installation</i> .						

4-4 WATER COOLING REQUIREMENTS

4-4-1 Gradient Coil Temporary Backup Water Cooling Recommendation

There are no options available to support temporary backup water cooling for the TRM Gradient Coil. Water cooling must be provided by the cooling cabinet (TSCC or TGWC) supplied with the system to prevent contamination/damage to the coil and for proper image quality.

4-4-2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet (MS5). The backup cooling design can utilize open loop city water only as temporary backup during loss of the closed loop water cooling from the TSCC or customer provided water cooling to the Shield/Cryo Cooler Compressor. Long term open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching that can take place which will eventually contribute to failure. Water system capacity must be selected to make sure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Cooler Compressor Cabinet, refer to Tables 4-4 and 4-5 and Illustration 4-1 for water cooling specifications.



Switching the Shield/Cryo Cooler Compressor inlet/outlet cooling from the TSCC to a temporary water backup supply will result in approximately 1.5 gallons (5.5 liters) of 50% mixture of Dowfrost HD and de-ionized water being discharged. This discharge may have site impacts due to local regulatory codes. Make sure to understand and follow local regulatory requirements when designing and implementing a temporary backup water system. The design of the change over equipment from TSCC to city water and vice-a-versa must not allow contamination of the closed loop system in the TSCC.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

Note

Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

TABLE 4-4
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.

4-4-2 Shield/Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation (Continued)

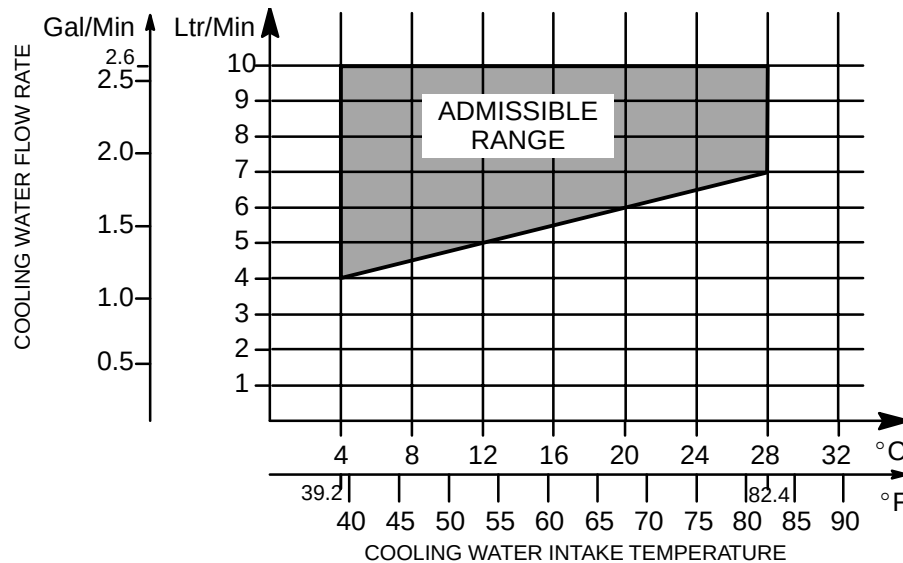
TABLE 4-5
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPERATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Note 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28)	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2 & 3	7.5 (0.5) [52] at minimum flow rate See Note 5	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
			Maximum 2.6 (10) See Notes 2 & 3	47 (3.2) [324] at maximum flow rate See Note 5	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The backup cooling system design must have allowances for pressure/temperature changes due to distance located from the Shield/Cryo Cooler Compressor.

- 1 Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.
- 2 Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4-1 for graphic water temperature and flow rate admissible range.
- 3 Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.
- 4 Water flow rate and temperature rise value are based on water.
- 5 Pressure drop values based on new system, may rise due to calcification.
- 6 Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.



SHIELD/CRYO COOLER COMPRESSOR REQUIREMENTS FOR TEMPORARY BACKUP WATER COOLING
ILLUSTRATION 4-1

4-4-3 Water Cooled TSCC Requirements

The water cooled configuration of the Twin System Cooling Cabinet (TSCC) contains a dedicated, closed loop, liquid-to-liquid water chiller system providing water cooling for the Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil. The water cooled TSCC configuration requires customer provided water cooling, refer to Table 4-6 and Illustration 4-2 for cooling water requirements. The water cooled TSCC has 1-1/2 inch (38 mm) NPT female connections at the lower back of the TSCC unit for customer water cooling connection. Refer to Illustration 2-17 for water supply and return connection location details. The water supply and return lines must be cleared of debris (solder, dirt, etc.) before connecting to the TSCC, refer to **APPENDIX C Prerequisites for Scheduled Start-up of Chillers For Signa TwinSpeed**.

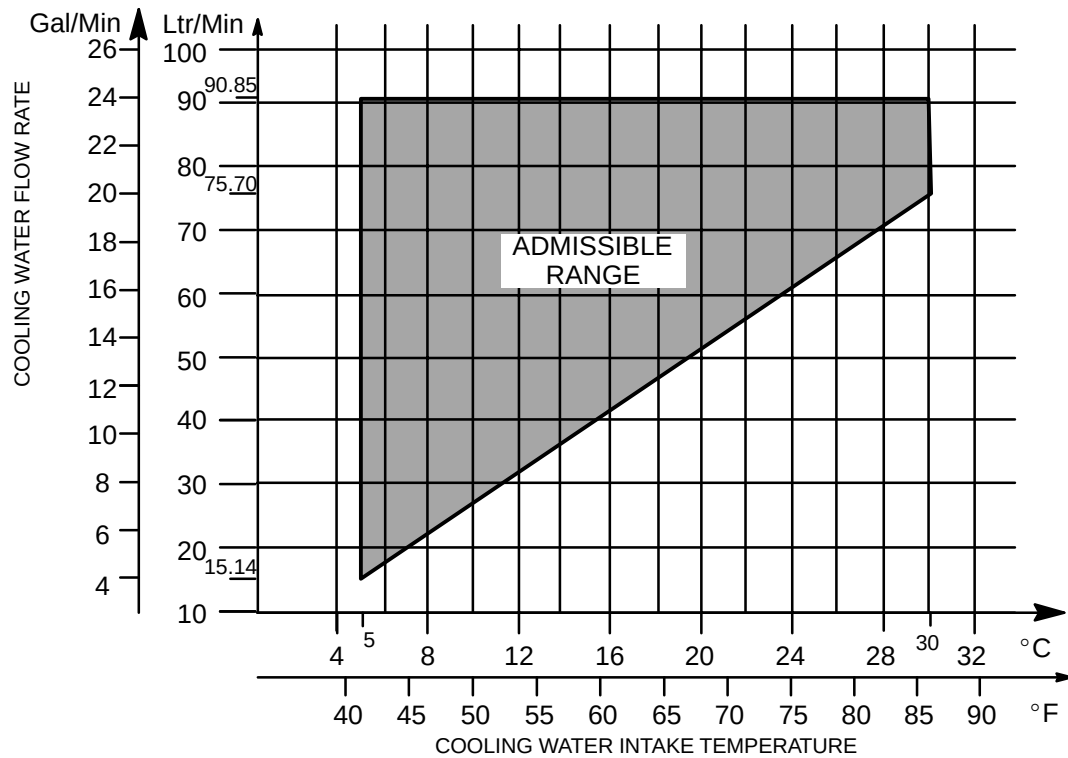
Note

Continuous water cooling is critical for the TSCC and Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week / 365 days per year to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

TABLE 4-6
WATER COOLED TSCC COOLING WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TSCC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TSCC filter at cabinet inlet.
Anti-freeze	Propylene Glycol (50% maximum concentration in water)	
Inlet Pressure	150 psi maximum	
Pressure Drop	1.6 psi (11 kPa) at minimum flowrate of 4 gpm (15.14 L/min) 5.6 psi (38.5 kPa) at maximum flowrate of 20 gpm (75.70 L/min)	
Maximum Heat Output	100,000 BTU/Hr 29,307 Watts to water 12,000 BTU/Hr 3517 Watts to air	Refer also to Section 4-3 AIR COOLING REQUIREMENTS

4-4-3 Water Cooled TSCC Requirements (Continued)



TSCC COOLING WATER FLOWRATE/TEMPERATURE REQUIREMENT
ILLUSTRATION 4-2

4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

The Shield/Cryo Cooler Compressor customer provided water cooling system must be closed loop design. The water cooling design can utilize open loop city water, with required filtering, only as temporary backup during loss of closed loop water cooling system. Open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching to take place which will eventually contribute to failure. Water system capacity must be selected to insure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Compressor.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week / 365 days per year to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

The Shield/Cryo Cooler Compressor closed loop system may be shared with other equipment in the MR suite. The number of sharing systems should be kept to as low as possible in order to minimize contamination and reliability problems. Flow gauges and valves are recommended at all branch lines to control distribution and allow servicing of equipment.

Refer to Tables 4-4 and 4-5 for water cooling specifications.

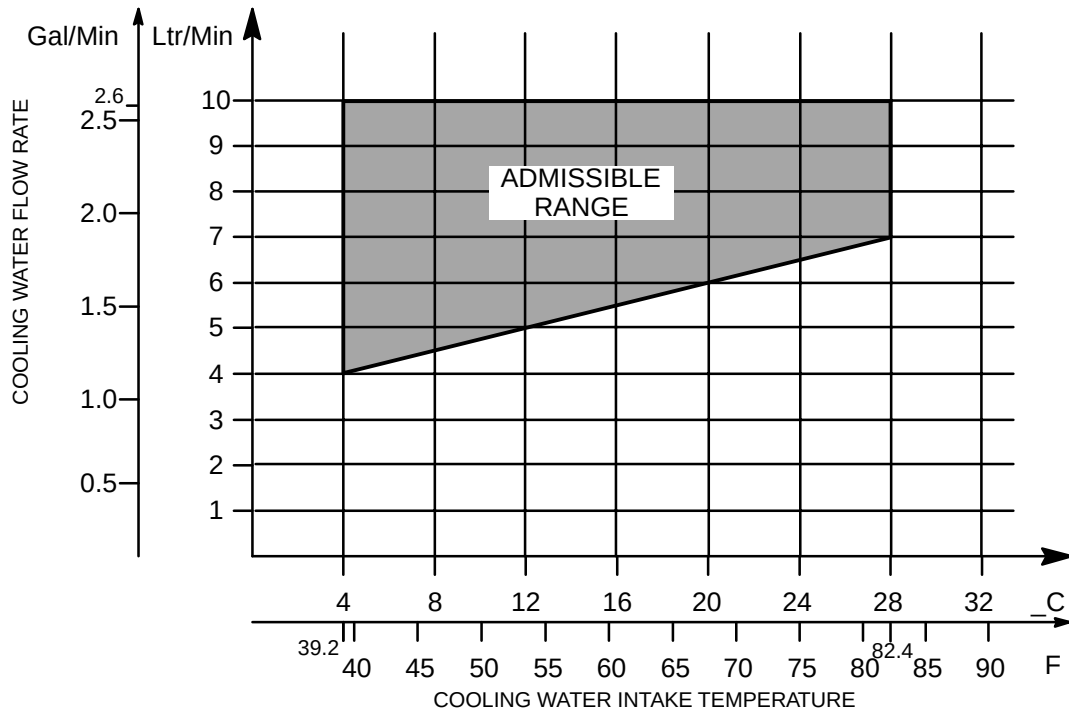
Note

The **Shield/Cryo Cooler Compressor** requires customer supplied flexible hose for mating with water cooling. Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

4-4-4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC (Continued)

TABLE 4-7
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.
Anti-freeze	Preferred minimum 25% by Volume Maximum 50% by volume	Use stabilized product to reduce corrosion and organic growth. Preferred minimum value to minimize organic growth.



SHIELD/CRYO COOLER COMPRESSOR COOLING WATER REQUIREMENT
ILLUSTRATION 4-3

4-4-4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TWGC (Continued)

TABLE 4-8
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPERATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Notes 3, 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28) See Note 9	Minimum 29 (200)	Minimum 1.1 (4) See Notes 2, 3 & 9	7.5 (0.5) [52] at minimum flow rate See Notes 2, 5, & 9	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
		Maximum 100 (690)	Maximum 2.6 (10) See Notes 2, 3 & 9	47 (3.2) [324] at maximum flow rate See Notes 2, 5, & 9	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The cooling system design must have allowances for pressure/temperature changes due to distance from the Shield/Cryo Cooler Compressor.

- Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.
- Water Flowmeter Kit (46-294052G1) is available to check/monitor flow rate for Shield/Cryo Cooler Compressor. Add 2 psi to total system pressure drop if flowmeter is permanently installed in system.
- Recommend a flowmeter be permanently installed in system, include flowmeter drop in total system pressure drop.
- Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4-1 for graphic water temperature and flow rate admissible range.
- Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.
- Water flow rate and temperature rise value are based on water. Laboratory grade Ethylene Glycol pr Propylene Glycol anti-freeze may be used (do not mix Ethylene Glycol with Propylene Glycol). Preferred concentration is 65% water and 35% Ethylene Glycol to minimize organic growth. Concentration of 50/50 is acceptable with a derate of 0.8 in specific heat calculations and a 20% increase in flow.
- Pressure drop values based on new system, may rise due to calcification.
- Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.
- Water Cooling Circuit typical values:
 - water inlet flow 1.8 to 2.1 gal/minute (7 to 8 liter/minute)
 - water inlet temperature 53.6 to 59°F (12 to 15°C)
 There is a risk of damaging the Shield/Cryo Cooler Compressor with water inlet low temperature and low flow range.

4-4-5 Water Cooled TGWC Requirements

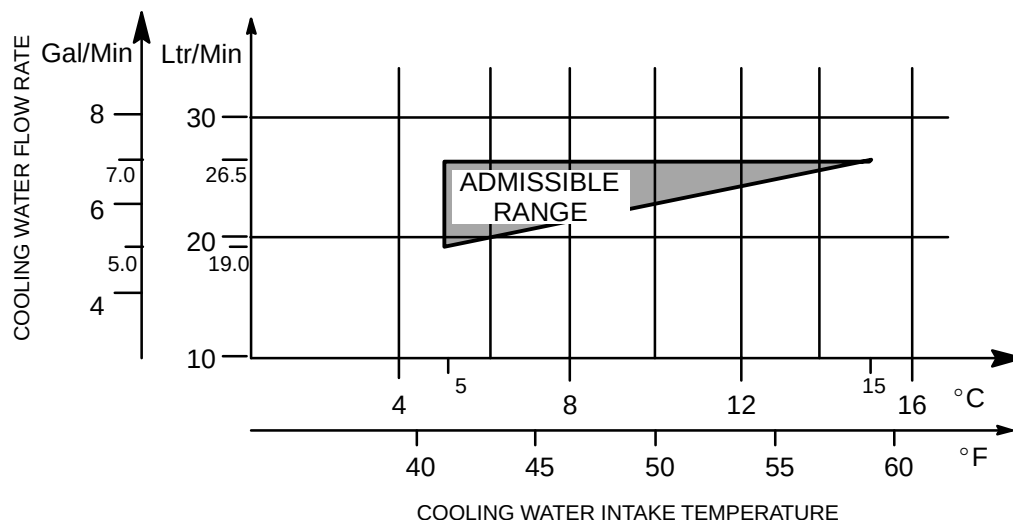
The Water Cooled configuration of the Twin Gradient Water Cooler (TGWC) contains a dedicated, closed loop, liquid-to-liquid heat exchanger system providing water cooling for the Gradient Coil ONLY. The Water Cooled TGWC configuration requires customer provided water cooling, refer to Table 4-9 for cooling water requirements. The water cooled TGWC has 0.5 inch (12.7 mm) female thread connections at the lower left side of the unit for customer water cooling connections. Refer to Illustration 2-24 for water supply and return connection location details.

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor. Refer to **Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC** for customer provided water cooling requirements.

TABLE 4-9
WATER COOLED TGWC WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
Composition	Water	Maximum of 50% glycol in water mixture
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TGWC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TGWC filter at cabinet inlet.
Maximum Heat Output	37,000 BTU/Hr 10,843 Watts	Refer also to Section 4-3 AIR COOLING REQUIREMENTS
Pressure Drop	9.02 PSIG (62.2 kPa) at minimum flowrate of 5.0 gpm (19.0 liters/minutes) 17.86 PSIG (123.1 kPa) at maximum flowrate of 7.0 gpm (26.5 liters/minutes)	These pressure drop values are for pure water is used. Multiply the values by 1.55 if 50/50 Propylene Glycol / water is used.
Inlet Pressure	150 psi maximum	



TGWC COOLING WATER FLOWRATE/TEMPERATURE REQUIREMENT

ILLUSTRATION 4-4

4-5 ALTITUDE

100 ft (30.5 m) below sea level to 11,808 ft (3600 m) above sea level.

4-6 LIGHTING

Direct Current (DC) lighting is recommended in the Magnet Room. Alternating Current (AC) lighting, other than fluorescent, may be used but the magnetic field of the system will significantly reduce the operating life of incandescent bulbs in AC lighting. An optional DC Lighting product is available which provides for two circuits of DC power and adjustable light levels in the Magnet Room. Refer to Table 4-10 for additional lighting requirements.

Note

Fluorescent lighting is not allowed in the Magnet Room due to the RF noise generated by the fluorescent light tubes.

TABLE 4-10
ROOM LIGHTING REQUIREMENTS

AREA	LIGHTING TYPE	LUMINOUS INTENSITY	NOTES
Magnet Room	Direct Current (DC) Incandescent or Quartz (See Notes)	• Minimum 300 lux around the front of the magnet for patient access. • Need provision to provide 300 lux above the magnet (non-magnetic, portable lighting is acceptable).	• Direct Current (DC) lighting is recommended in the Magnet Room to minimize light bulb life issues. • Electrical lines for AC lighting must meet filtering and grounding requirements in Section 7-4 ELECTRICAL. • Fluorescent lighting is not allowed in the Magnet Room to avoid RF noise. • Short filament length is recommended, linear lamps are not recommended because of the filament length and high incidence of filament failure. • The alternating current (AC) ripple from the DC power should be no greater than 5%. • Dimmers in the Magnet Room are not acceptable. If a low and high light level is desired, the different levels must be selectable by a switch.

4-7 ACOUSTICS

The following acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR-generated acoustic levels, see the system operator manual.

4-7-1 Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics of the room where patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structure borne paths.

Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustics issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

Note

Less than 5% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustics issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

4-7-2 System Acoustic Noise Levels

Any GE factory-installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under **Operating Conditions** Condition 1 below. Typical scans generate acoustic levels as stated under **Operating Conditions** Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, hearing protection is required for all people in the Magnet Room during scans to prevent hearing impairment, acoustic levels may exceed 99dBA. Again, for more information about recommended safety procedures regarding patient exposure to MR-generated acoustic noise, see the MR system Operator Manual.

Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area 55 dBA
- Equipment Room 75 dBA
- Outdoor TSCC (Twin System Cooling Cabinet) 75 dBA
- Outdoor TGWC (Twin Gradient Water Cooler) 75 dBA

Operating Conditions

Condition 1

MR scanners under “worst-case” operating conditions, could generate acoustic levels (as measured at the magnet iso-center) as follows:

Average SPL 115 dBA SPL= Sound Pressure Level
 Peak 120 dB
 Frequency Range 20 to 20k Hz

Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet iso-center) somewhat lower as follows:

Average SPL 85 to 95 dBA
 Peak 105 to 112 dB
 Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the “worst case” Condition 1, mentioned above. Note that high-field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

4-7-3 Design Guidelines

Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to Section 4-7-2 System Acoustic Noise Levels. The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T-bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

Inter-Spacial Areas

Acoustic Noise Control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

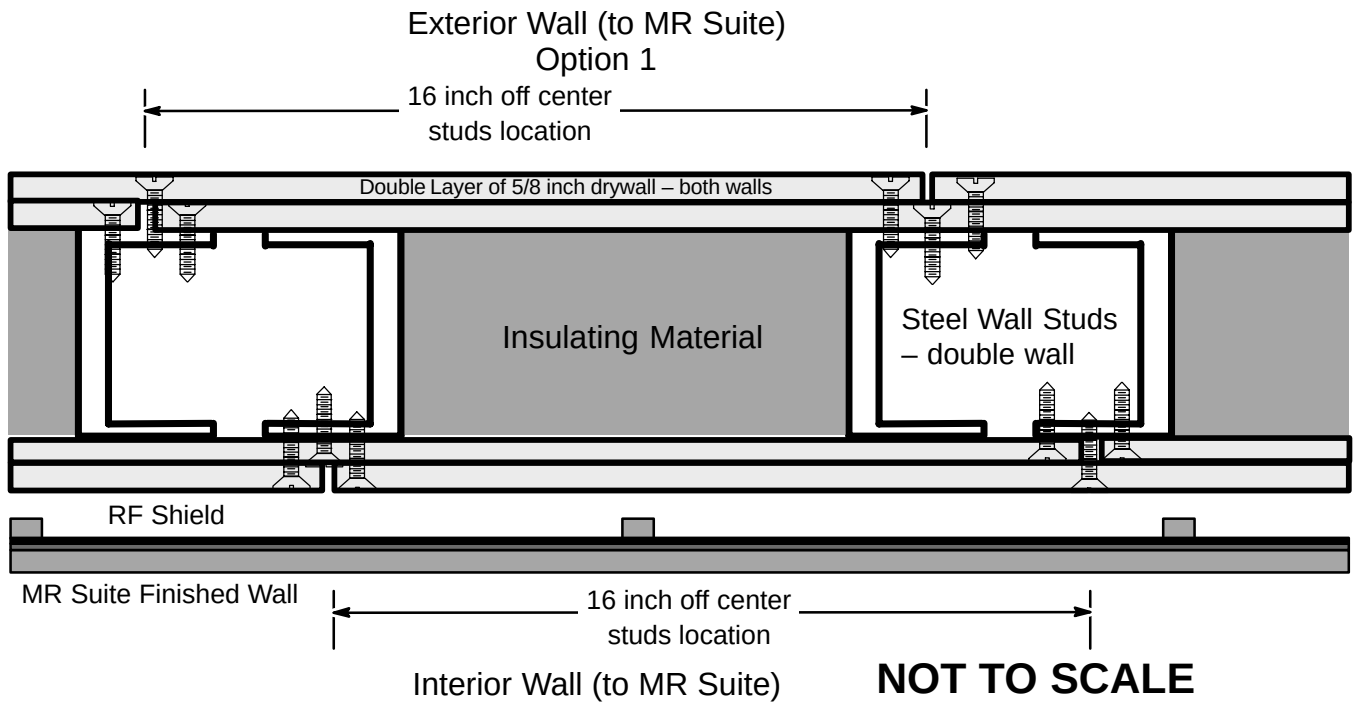
The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

Wall Construction

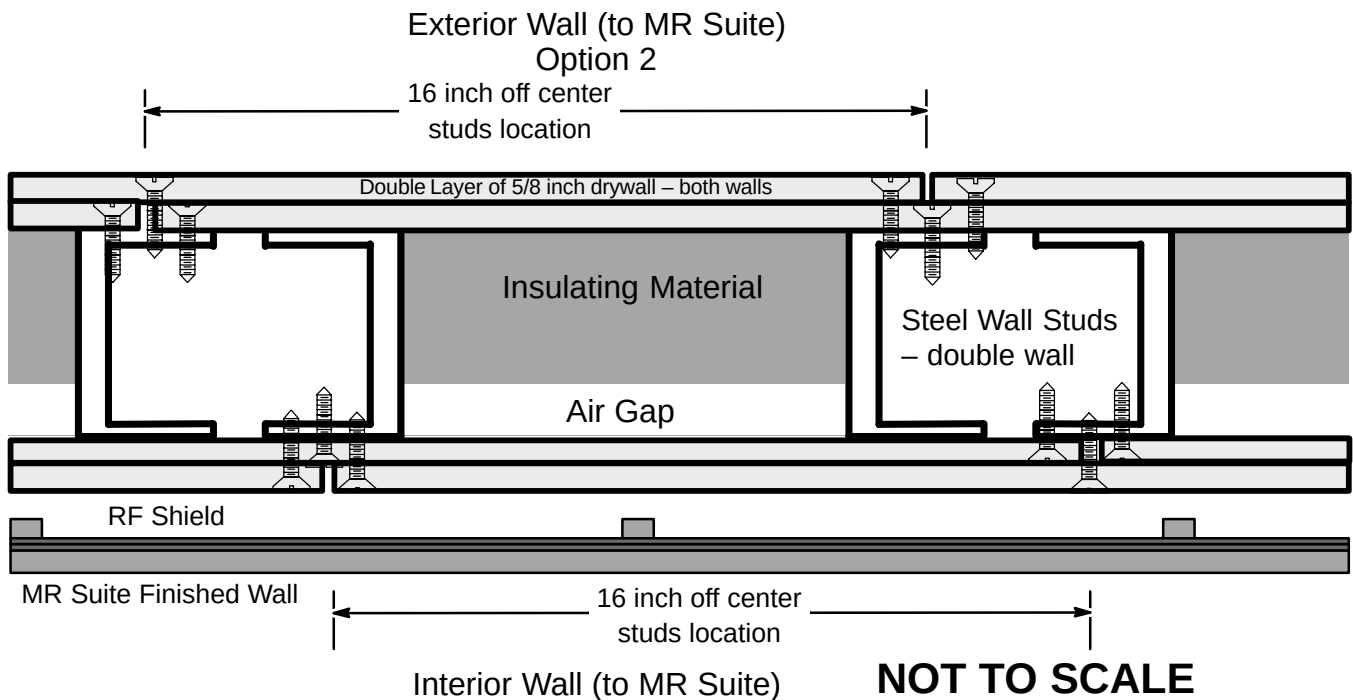
Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3-5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non-hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non-hardening) and fiberglass batt material. See examples of wall construction shown in Illustrations 4-5 and 4-6.
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non-hardening) will be used to seal the remaining cracks and gaps.

4-7-3 Design Guidelines (Continued)



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 1
ILLUSTRATION 4-5



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 2
ILLUSTRATION 4-6

4-7-3 Design Guidelines (Continued)

High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall "doghouse".

Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

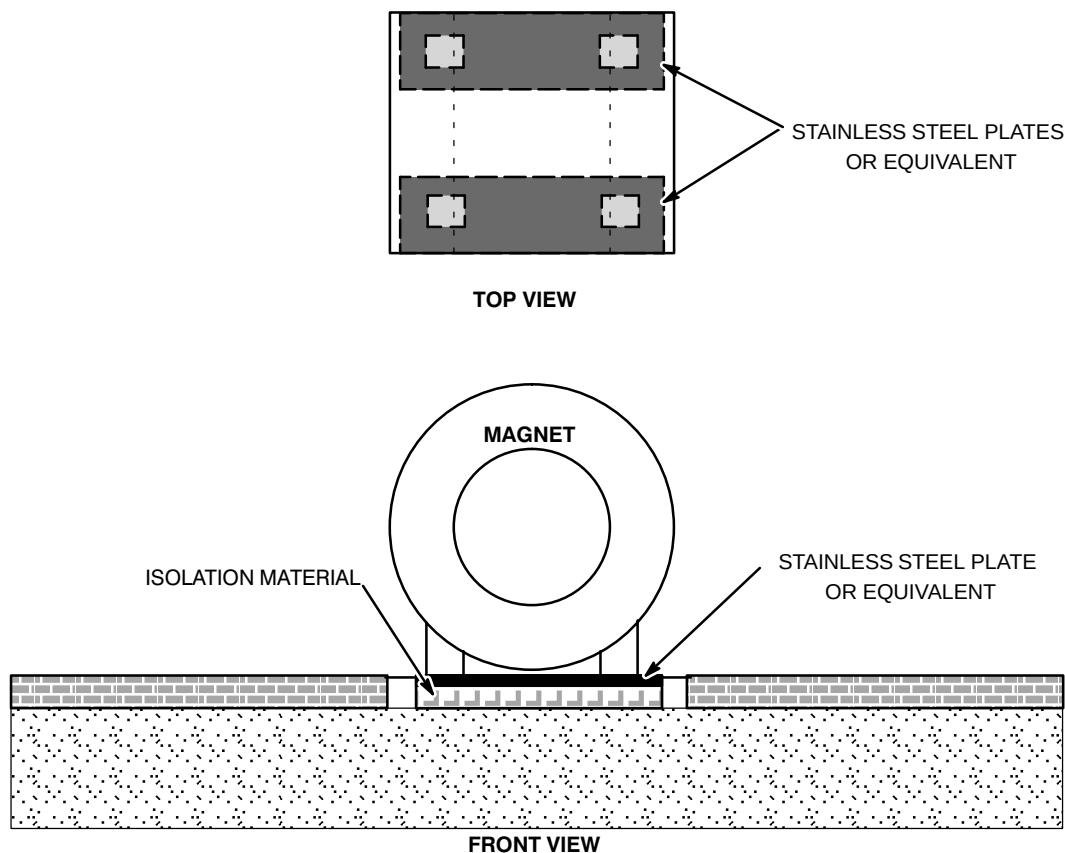
- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.
- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 50 to 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide an STC 50 to 60 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

4-7-3 Design Guidelines (Continued)

Structureborne Vibration Control

Upper floor MR installations represent the largest population of sites which may experience structureborne acoustic issues. Typically, an architect, acoustic engineer, or project engineer would suggest placing the magnet on a (soft) vibration isolator. Although this classical solution may reduce the magnet generated vibration from transmitting into the building structure, this solution has produced mixed results ranging from introducing image quality issues with proper vibration attenuation to not providing enough vibration attenuation.

GE has worked with consultants who have designed/installed solution that provide (large) vibration attenuation without compromising image quality. Such solutions would bolt the magnet to stiff plates to meet the magnet bolting requirement (refer to **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM**). The plate and isolation material would be selected/sized to distribute the vibration over a large surface area and the isolation system would be selected to sufficiently attenuate vibration from the building structure. Additionally the solution natural frequency misaligns with the floor resonance. The solutions used typically had higher natural frequency than the supporting floor. Note: the magnet plate and selected isolation must ensure the magnet vertical, horizontal, and pitch/yaw modes do not compromise the vibration specification or floor levelness requirements, refer to **Section 4-15 VIBRATION** and **Section 7-6-4 Floors**. See Illustration 4-7 for an overview of such a solution concept.



OVERVIEW STRUCTUREBORNE VIBRATION CONTROL SOLUTION CONCEPT

ILLUSTRATION 4-7

4-8 ROOM VENTILATION

Refer to Table 4-11 for ventilation specifications for the magnet and cryogen storage rooms. Refer to Section 4-12, POLLUTION, for air quality specifications.

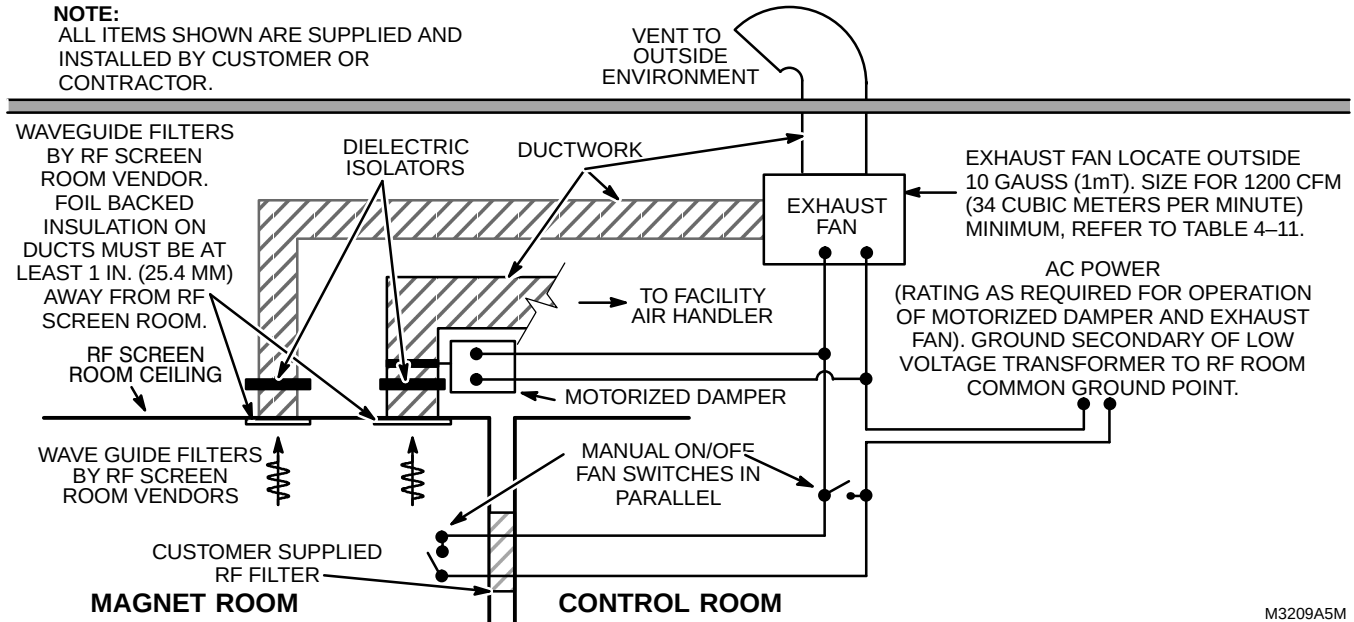
TABLE 4-11
ROOM VENTILATION REQUIREMENTS/RECOMMENDATIONS

ROOM	VENTILATION REQUIREMENTS	VENTILATION RECOMMENDATIONS
MAGNET	• Sufficient air ventilation in the magnet room must be maintained, not only for patient comfort during scans but also to maintain proper oxygen level during cryogen replenishment. • An exhaust fan to be placed above RF shielding with appropriate wave guide filtering for quick removal of helium gas if large amounts of helium disperse into magnet room. Inert gas containers, such as dewars, are not air tight. • Magnet Room exhaust fan intake vent must be located at the highest ceiling plane near the magnet cryogen vent. • Exhaust fan to exhaust to safe outside area and be independent of cryogenic venting. • The exhaust fan and air inlet must be sized for an air flow of 1200 CFM (34 m³/minute) minimum. • Two manual exhaust fan controls connected parallel, one to be located near the Operator Console and second control located in the Magnet Room. Refer to Illustration 4-8 for exhaust fan recommended set-up or Illustration 4-9 for recommended set-up with optional Oxygen Monitor. • Exhaust fan (customer supplied) to be installed and operating before magnet is moved into room. • Provide minimum 2 ft x 2 ft (0.61 m x 0.61 m) pressure equalizing waveguide vent in the magnet room ceiling to prevent positive or negative pressures from interfering with opening of the magnet room door per IEC 60601-2-33 6.8.3 cc.	• Minimum 5-7% of outside makeup air to be vented into magnet room. For example, with an air input rate of 1200 cubic feet per minute (CFM) (34 cubic meters per minute), there must be a minimum of 60 CFM (1.7 cubic meters per minute) (5%) of outside makeup air. • The exhaust fan and air inlet provide for a minimum of 12 air exchanges per hour or 1200 CFM (34 m³/minute), which ever is larger. • Note: Annual customer inspection and cleaning / maintenance of the exhaust fan system (fan, inlet grill/filter, ducts, etc.) is needed to meet the minimum airflow requirement to an outside area.
CRYOGEN STORAGE	• Adequate ventilation to ensure acceptable oxygen level per local regulation.	

4-8 ROOM VENTILATION (Continued)

NOTE:

ALL ITEMS SHOWN ARE SUPPLIED AND INSTALLED BY CUSTOMER OR CONTRACTOR.

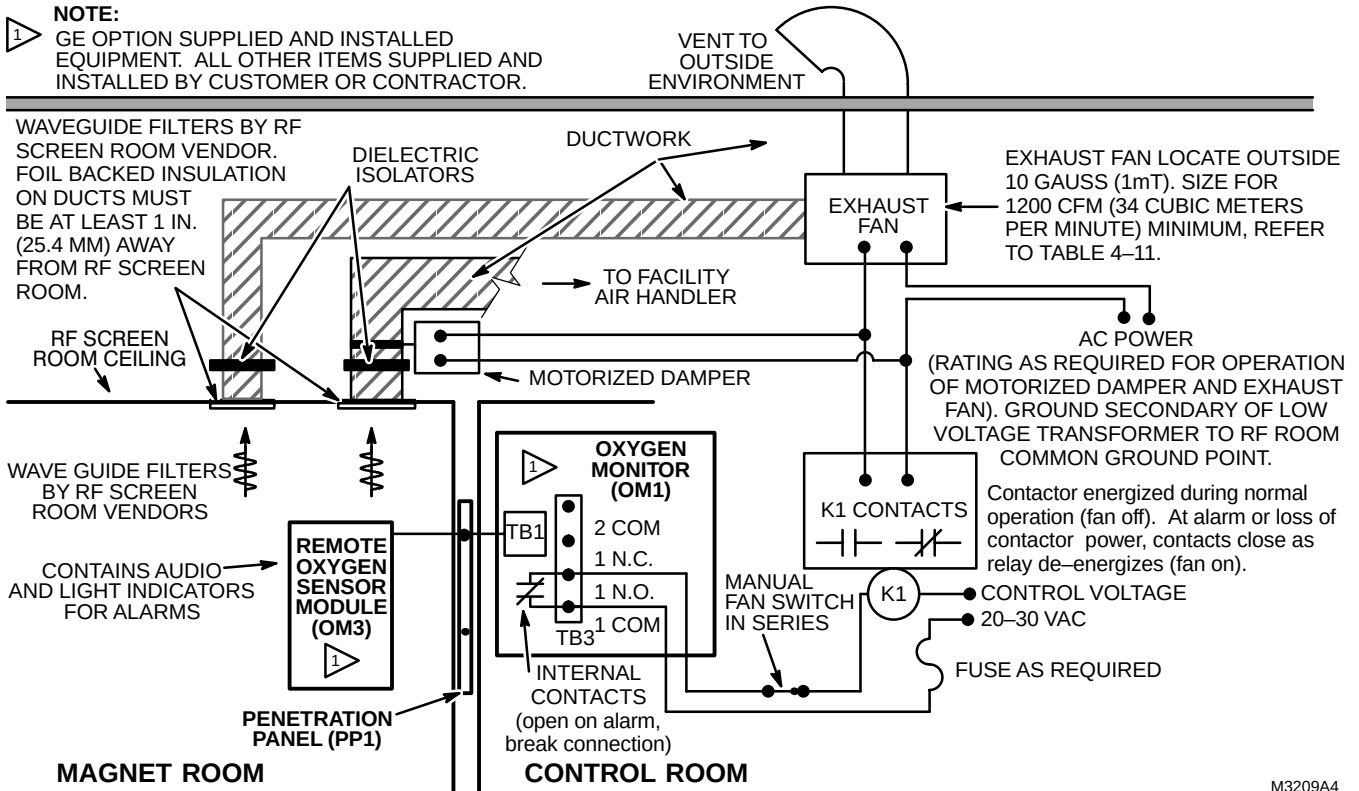


M3209A5M

EXHAUST FAN SET-UP
ILLUSTRATION 4-8

NOTE:

GE OPTION SUPPLIED AND INSTALLED EQUIPMENT. ALL OTHER ITEMS SUPPLIED AND INSTALLED BY CUSTOMER OR CONTRACTOR.



M3209A4

EXHAUST FAN SET-UP WITH OPTIONAL OXYGEN MONITOR
ILLUSTRATION 4-9

4-9 CRYOGENIC VENTING

The superconducting magnet used in the MR System contains large amount of liquid helium at 4 Kelvin (K) = -452° Fahrenheit (F) or -269° Celsius (C). During a quench, the magnet quickly boils off 90% of this liquid. Consequently, a very large volume of extremely cold helium gas must be safely vented outside of the building. Failure of the cryogenic vent can result in this cold gas entering the magnet room or another portion of the building and lowering the ambient oxygen supply to an unsafe level. Therefore it is very important that designer/contractors familiar with industrial piping systems be responsible for the vent system design and installation.

When a quench occurs, the helium gas will immediately begin to warm up and expand rapidly. As the gas escapes from the magnet, the helium gas will increase in temperature from 4.5 K to approximately 30 K. As a result, the gas will expand to 10 times of its original volume. Consequently, venting systems must never decrease in size and may increase in size (diameter) as the distance from the magnet increases.



THE CRYOGENIC VENT INSPECTION MUST BE COMPLETED TO FINAL EXIT OUTSIDE OF THE BUILDING INCLUDING RF SHIELD PENETRATION PRIOR TO MOVING THE MAGNET INTO THE MAGNET ROOM.

IN THE SITUATION WHERE THE MAGNET DELIVERY WILL BE THROUGH A ROOF HATCH AND THE CRYOGENIC VENT WILL BE LOCATED IN THE SAME HATCH THEN THE CRYOGENIC VENT MUST BE INSTALLED AND INSPECTED TO FINAL EXIT WITHIN 24 HOURS OF THE MAGNET DELIVERY THROUGH THE HATCH.

Note

To minimize the confusion due to various domestic and foreign venting material sizing systems, the actual dimensions of the vent material are used to describe the required vent sizes in this document. Please use the locally available venting material best matching or exceeding the requirements discussed in this document.

Note

The difference in the American terminologies of pipes and tubes is disregarded in this document.

Note

The vent size is described by the inside diameter in the pressure drop tables, because the pressure drop is calculated with the inside diameter. The vent size is described by the outside diameter for the waveguide requirements, because sealing the Ventglas[®] and matching the outside diameters are the critical issues.

Note

Portions of this document may not be applicable to the mobile units. Cryogenic vent design for mobile units must be submitted to GE Medical Systems MR Engineering for certification.

4-9-1 Requirements For Outside Magnet Room

The customer's vent system designer/contractor must design and install the cryogenic vent from the RF shield waveguide to the final exit on the building roof top or outside wall. Table 4-12 contains the requirements which must be strictly adhered to in order to prevent failure of the vent system during a magnet quench.

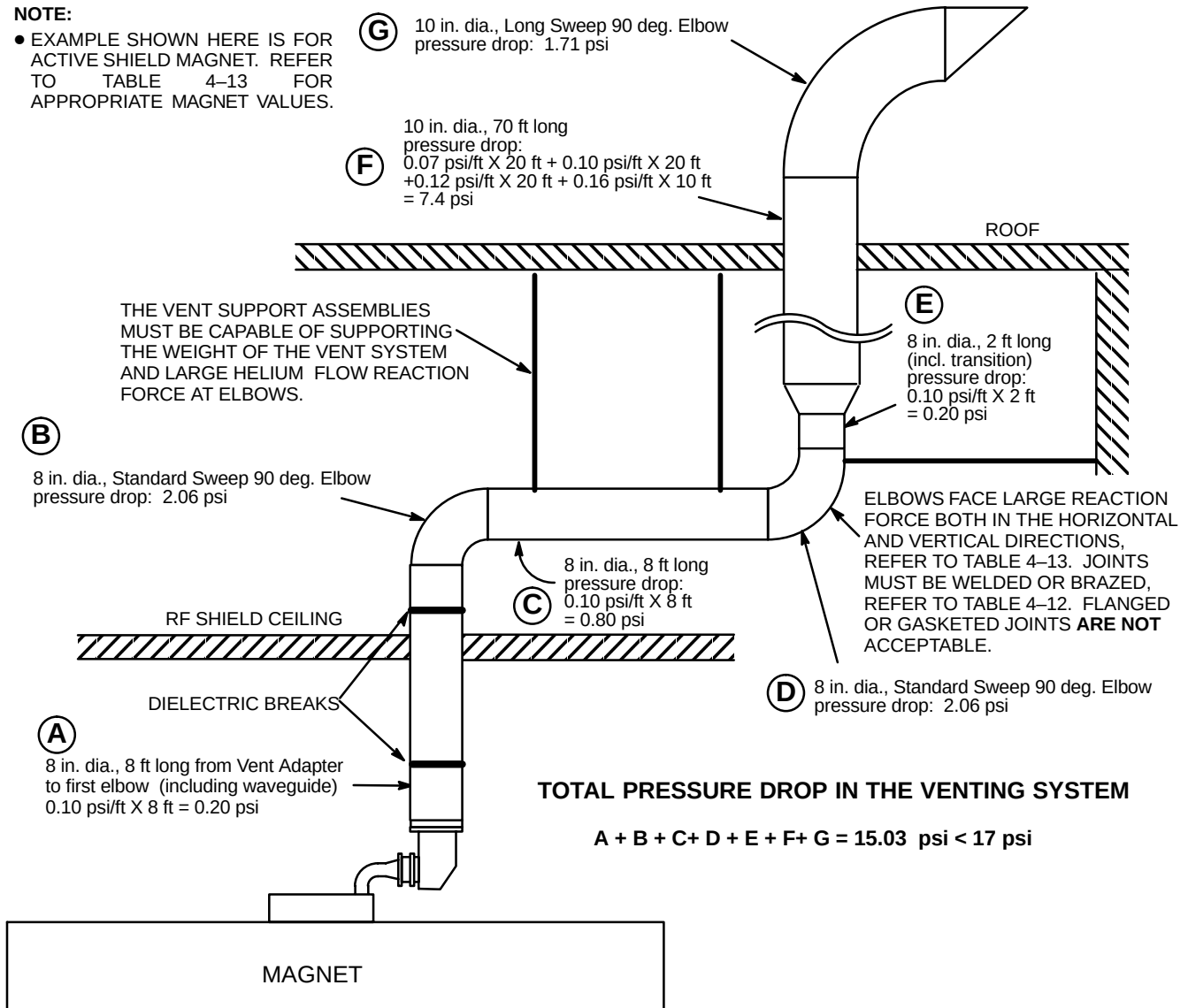
TABLE 4-12
CRYOGENIC VENTING REQUIREMENTS FOR OUTSIDE MAGNET ROOM

PARAMETER	CRYOGENIC VENTING REQUIREMENTS
VENT SIZE	- The total pressure drop of the entire cryogenic vent system must be less than 17 psi (117.2 KPa). The calculation starts at the magnet vent interface and ends at the termination point outside of the building. Use Table 4-13 to determine proper vent sizes. See Note 1. See Illustration 4-10 for a sample calculation. - The pressure drop of the RF shield waveguide must be included in the overall calculation. Make sure the RF shielded room supplier provides a straight pipe for the waveguide with an outside diameter which matches the tubing from the magnet within ± 0.125 inch (3 mm). Refer to Section 7, RF SHIELDED ROOM, for location of vent pipe/waveguide. - The vent route must be as direct as possible and designed with a minimum number of elbows. If elbows must be used, they must be standard or long sweep types with K factors as shown in the footnotes of Table 4-13. Use of elbows with higher K factors will result in higher pressure drops than those shown in Table 4-13. Refer to Section 7, RF SHIELDED ROOM, for routing of vent pipe.
VENT CONSTRUCTION	- Expansion/contraction joints must be provided to account for dimensional changes that will occur during a magnet quench when the vent temperature decreases from ambient to 4.5 K (-451° F or -268° C). - At the waveguide, non-metallic isolation joint must be provided to ensure the integrity of the magnet room RF shield. The joint gap must be 1.0 ± 0.25 inch (25 ± 6 mm). - All components must be able to withstand the minimum static pressure as calculated from Table 4-13 and a helium flow reaction force at temperatures from 4.5 K (-451° F or -268° C) and above. - Electro-mechanical fire dampers must not be used in the design of the vent system. Fusible link fire dampers are acceptable and require routine inspection and maintenance. - Appropriate protection must be provided for any portion of the cryogenic vent system which may drip condensation on personnel or ceiling components. - Access must be available for any portions of the cryogenic vent system made of non-metallic materials which require routine inspection and maintenance by the customer.
VENT SUPPORT	- All portions of the cryogenic vent system must be adequately supported. The vent support assemblies must be capable of supporting the weight of the vent system and the helium flow reaction force of 1850 lbs (8229 N) for 1.5T magnet at the vent elbows. - Non-metallic (i.e Ventglas) joints must not be used as the support for the vent system.
VENT EXIT AND TERMINATION	WARNING! - The exit location of the cryogenic vent system must be chosen to prevent the extremely cold exhaust gas from injuring anyone, including maintenance personnel. Exhaust opening access must be limited by warning signs or other barriers within a distance 20 feet (6.1 m) for 1.5T LCC (Active Shield) magnets. - The exhaust vent must be directed in a manner to prevent the cold gas from injuring personnel or damaging any building components. It must also be directed away from air intake vents. - For a roof top exit, GE recommends that the exhaust flow be directed horizontally using a 90° elbow having minimal pressure drop and the outlet covered with a 0.5 inch (12.7 mm) mesh screen to prevent the entry of foreign material. See Illustration 4-11. Other low pressure drop, high flow rate roof caps are acceptable. - The bottom of the 90° elbow must be at least 3 feet (0.9 meters) above the roof deck. This dimension must be higher if the location of the vent exhaust is susceptible to being blocked by drifting snow.
Note: 1 The customer's designer is responsible for selecting materials and hardware capable of safely handling the pressures and cold temperature generated within the vent at each MRI site.	

4-9-1 Requirements For Outside Magnet Room (Continued)

NOTE:

- EXAMPLE SHOWN HERE IS FOR ACTIVE SHIELD MAGNET. REFER TO TABLE 4-13 FOR APPROPRIATE MAGNET VALUES.



TOTAL PRESSURE DROP IN THE VENTING SYSTEM

$$A + B + C + D + E + F + G = 15.03 \text{ psi} < 17 \text{ psi}$$

SAMPLE PRESSURE DROP CALCULATION OF CRYOGENIC VENTING

ILLUSTRATION 4-10

4-9-1 Requirements For Outside Magnet Room (Continued)

TABLE 4-13
CRYOGENIC VENT SYSTEM PRESSURE DROP MATRIX FOR 1.5 T LCC (ACTIVE SHIELD) MAGNET
(THIS TABLE MUST BE USED FOR CRYOGENIC VENT SYSTEM DESIGN)

CRYOGENIC VENT SYSTEM PRESSURE DROP MATRIX FOR A 1.5 TESLA LCC (ACTIVE SHIELD) MAGNET				PRESSURE DROP PER ELBOW USED ANYWHERE WITHIN A 20 FT VENT SEGMENT			
INSIDE DIAMETER OF VENT PIPE in. (mm)	DISTANCE OF VENT SYSTEM COMPONENT FROM MAGNET		PRESSURE DROP FOR STRAIGHT VENT PIPE WITH SMOOTH INSIDE SURFACE	STANDARD SWEEP 45° ELBOW	STANDARD SWEEP 90° ELBOW	LONG SWEEP 45° ELBOW	LONG SWEEP 90° ELBOW
	ft	(m)					
			psi/ft (KPa/m)	psi (KPa)	psi (KPa)	psi (KPa)	psi (KPa)
8 (203.2)	0-20	(0-6.1)	0.10 (2.26)	1.10 (7.58)	2.06 (14.20)	0.55 (3.79)	1.03 (7.10)
	20-40	(6.1-12.2)	0.21 (4.75)	2.10 (14.48)	3.70 (25.51)	1.03 (7.10)	1.85 (12.76)
	40-60	(12.2-18.3)	0.30 (6.79)	2.88 (19.86)	5.21 (35.92)	1.44 (9.93)	2.60 (17.92)
	60-80	(18.3-24.4)	0.38 (8.60)	3.70 (25.51)	6.71 (46.27)	1.85 (12.76)	3.36 (23.17)
	80-100	(24.4-30.5)	0.47 (10.63)	4.52 (31.17)	8.22 (56.68)	2.26 (15.58)	4.11 (28.34)
10 (254.0)	0-20	(0-6.1)	0.03 (0.68)	0.55 (3.79)	0.82 (5.65)	0.27 (1.86)	0.41 (2.83)
	20-40	(6.1-12.2)	0.07 (1.58)	0.82 (5.65)	1.51 (10.41)	0.41 (2.83)	0.75 (5.17)
	40-60	(12.2-18.3)	0.10 (2.26)	1.23 (8.48)	2.19 (15.10)	0.62 (4.27)	1.10 (7.58)
	60-80	(18.3-24.4)	0.12 (2.71)	1.51 (10.41)	2.74 (18.89)	0.75 (5.17)	1.37 (9.45)
	80-100	(24.4-30.5)	0.16 (3.62)	1.92 (13.24)	3.43 (23.65)	0.96 (6.62)	1.71 (11.79)
12 (304.8)	0-20	(0-6.1)	0.013 (0.29)	0.27 (1.86)	0.41 (2.83)	0.14 (0.97)	0.21 (1.45)
	20-40	(6.1-12.2)	0.027 (0.61)	0.41 (2.83)	0.82 (5.65)	0.21 (1.45)	0.41 (2.83)
	40-60	(12.2-18.3)	0.041 (0.93)	0.55 (3.79)	1.10 (7.58)	0.27 (1.86)	0.55 (3.79)
	60-80	(18.3-24.4)	0.054 (1.22)	0.69 (4.76)	1.37 (9.45)	0.34 (2.34)	0.69 (4.76)
	80-100	(24.4-30.5)	0.069 (1.56)	0.96 (6.62)	1.51 (10.41)	0.48 (3.31)	0.75 (5.17)
	100-120	(30.5-36.6)	0.08 (1.81)	1.09 (7.52)	1.77 (12.20)	0.55 (3.79)	0.88 (6.07)
	120-140	(36.6-42.7)	0.10 (2.26)	1.27 (8.76)	2.07 (14.27)	0.63 (4.34)	1.04 (7.17)
	140-160	(42.7-48.8)	0.11 (2.49)	1.43 (9.86)	2.36 (16.27)	0.72 (4.96)	1.19 (8.20)
	160-180	(48.8-54.9)	0.12 (2.71)	1.60 (11.03)	2.53 (17.44)	0.80 (5.52)	1.27 (8.76)
	180-200	(54.9-61.0)	0.17 (3.85)	1.75 (12.07)	2.93 (20.20)	0.88 (6.07)	1.47 (10.14)
14 (355.6)	0-20	(0-6.1)	0.008 (0.055)	0.20 (1.38)	0.301 (2.08)	0.102 (0.70)	0.15 (1.03)
	20-40	(6.1-12.2)	0.017 (0.12)	0.30 (2.07)	0.602 (4.15)	0.154 (1.06)	0.30 (2.07)
	40-60	(12.2-18.3)	0.026 (0.18)	0.40 (2.76)	0.808 (5.57)	0.198 (1.37)	0.40 (2.76)
	60-80	(18.3-24.4)	0.034 (0.23)	0.51 (3.52)	1.01 (6.96)	0.250 (1.72)	0.51 (3.52)
	80-100	(24.4-30.5)	0.043 (0.30)	0.71 (4.90)	1.11 (7.65)	0.353 (2.43)	0.55 (3.79)
	100-120	(30.5-36.6)	0.050 (0.34)	0.80 (5.52)	1.30 (8.96)	0.40 (2.76)	0.64 (4.41)
	120-140	(36.6-42.7)	0.063 (0.43)	0.933 (6.43)	1.52 (10.48)	0.46 (3.17)	0.76 (5.24)
	140-160	(42.7-48.8)	0.069 (0.48)	1.05 (7.24)	1.73 (11.93)	0.52 (3.59)	0.87 (6.00)
	160-180	(48.8-54.9)	0.076 (0.52)	1.18 (8.14)	1.85 (12.76)	0.59 (4.07)	0.93 (6.41)
	180-200	(54.9-61.0)	0.11 (0.76)	1.29 (8.89)	2.15 (14.82)	0.64 (4.41)	1.08 (7.45)
16 (406.4)	0-20	(0-6.1)	0.0053 (0.037)	0.153 (1.05)	0.230 (1.59)	0.078 (0.54)	0.115 (0.79)
	20-40	(6.1-12.2)	0.013 (0.09)	0.229 (1.58)	0.460 (3.17)	0.118 (0.81)	0.229 (1.58)
	40-60	(12.2-18.3)	0.020 (0.14)	0.306 (2.11)	0.618 (4.26)	0.152 (1.05)	0.306 (2.11)
	60-80	(18.3-24.4)	0.026 (0.18)	0.390 (2.69)	0.773 (5.33)	0.191 (1.32)	0.390 (2.69)
	80-100	(24.4-30.5)	0.033 (0.23)	0.543 (3.74)	0.850 (5.86)	0.270 (1.86)	0.421 (2.90)
	100-120	(30.5-36.6)	0.038 (0.26)	0.613 (4.23)	0.995 (6.86)	0.310 (2.14)	0.490 (3.38)
	120-140	(36.6-42.7)	0.048 (0.33)	0.714 (4.92)	1.16 (8.00)	0.352 (2.43)	0.581 (4.01)
	140-160	(42.7-48.8)	0.052 (0.36)	0.803 (5.54)	1.32 (9.10)	0.398 (2.74)	0.666 (4.59)
	160-180	(48.8-54.9)	0.058 (0.40)	0.903 (6.23)	1.42 (9.79)	0.451 (3.11)	0.712 (4.91)
	180-200	(54.9-61.0)	0.084 (0.56)	0.987 (6.81)	1.64 (11.31)	0.490 (3.38)	0.826 (5.70)

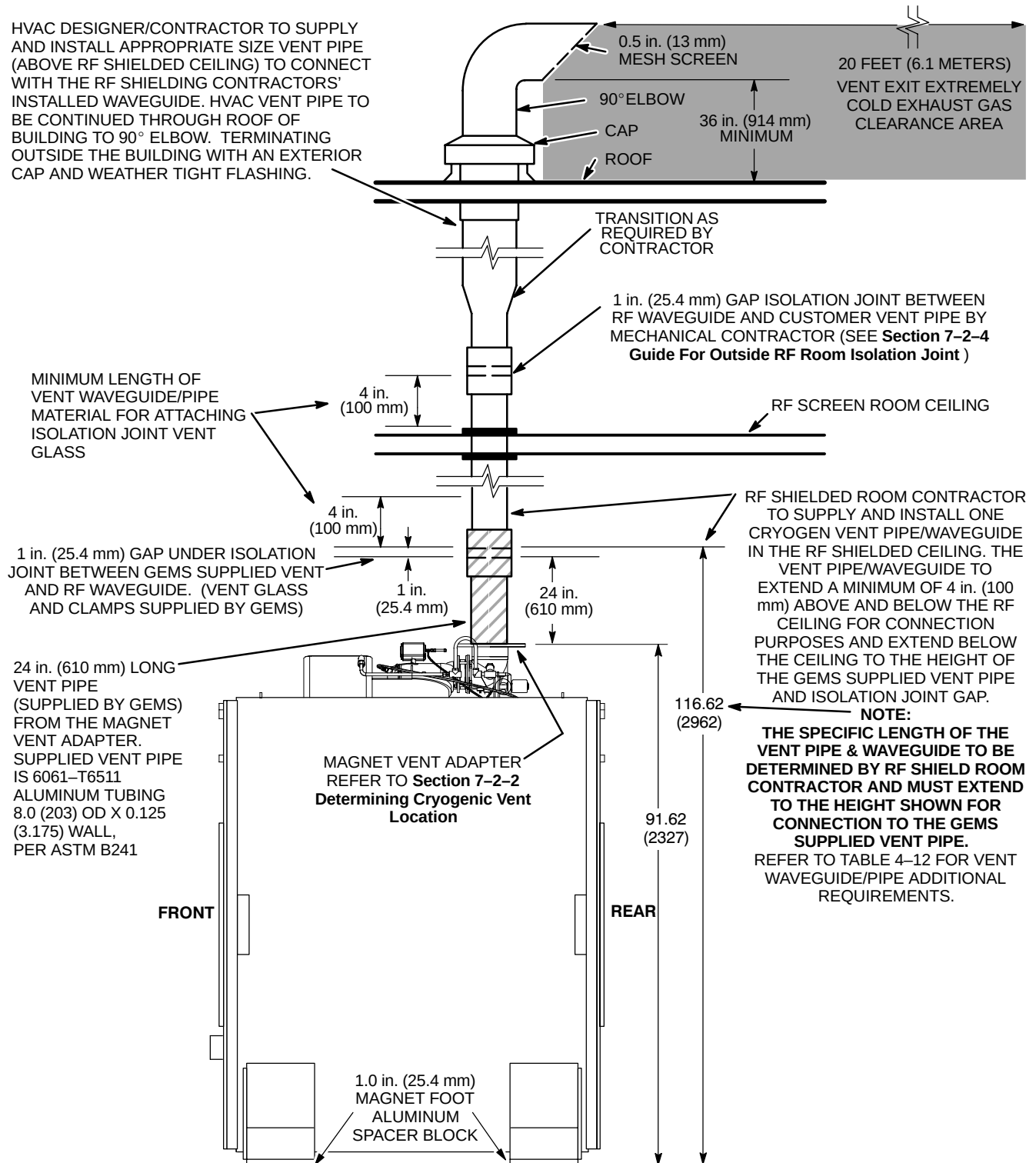
Note 1 : Elbows with angles greater than 90° **must not** be used.

Note 2 : The table data is based on the following:

- Initial flow conditions at magnet interface
- Gas temperature starting at 4.5 Kelvin (-452° F or -268° C).
- Helium gas flow rate of 2,737 cubic feet per minute (77.5 cubic meters per minute)
- 45° standard sweep elbow K = 15 F_t
- 90° standard sweep elbow K = 30 F_t
- 45° long sweep elbow K = 7.5 F_t
- 90° long sweep elbow K = 15 F_t

4-9-1 Requirements For Outside Magnet Room (Continued)

HVAC DESIGNER/CONTRACTOR TO SUPPLY AND INSTALL APPROPRIATE SIZE VENT PIPE (ABOVE RF SHIELDED CEILING) TO CONNECT WITH THE RF SHIELDING CONTRACTORS' INSTALLED WAVEGUIDE. HVAC VENT PIPE TO BE CONTINUED THROUGH ROOF OF BUILDING TO 90° ELBOW. TERMINATING OUTSIDE THE BUILDING WITH AN EXTERIOR CAP AND WEATHER TIGHT FLASHING.



TYPICAL CRYOGENIC VENT DETAIL

ILLUSTRATION 4-11

4-9-2 Requirements For Inside Magnet Room

GE provides and installs a vertical 8 in. (203.2 mm) outside diameter 24 in. (610 mm) long cryogenic vent tube within the magnet room straight up from the magnet in line with the waveguide in the RF Shield. The customer is responsible to provide additional vent tube within the magnet room to meet requirements defined in Table 4-14. The Vent diameter is determined by magnet, refer to Section 7-2-2, Determining Cryogenic Vent Location. For other vent configurations (i.e. offset ceiling exits, wall exits and geodesic domes) the customer's contractor is responsible for the design and installation of the cryogenic vent system and vent supports within the magnet room. In these cases, a complete description of the vent size, materials/properties and routing must be sent to GE Medical Systems MR Siting & Shielding Group for final review. Table 4-14 contains the GE requirements for vent design within the magnet room.

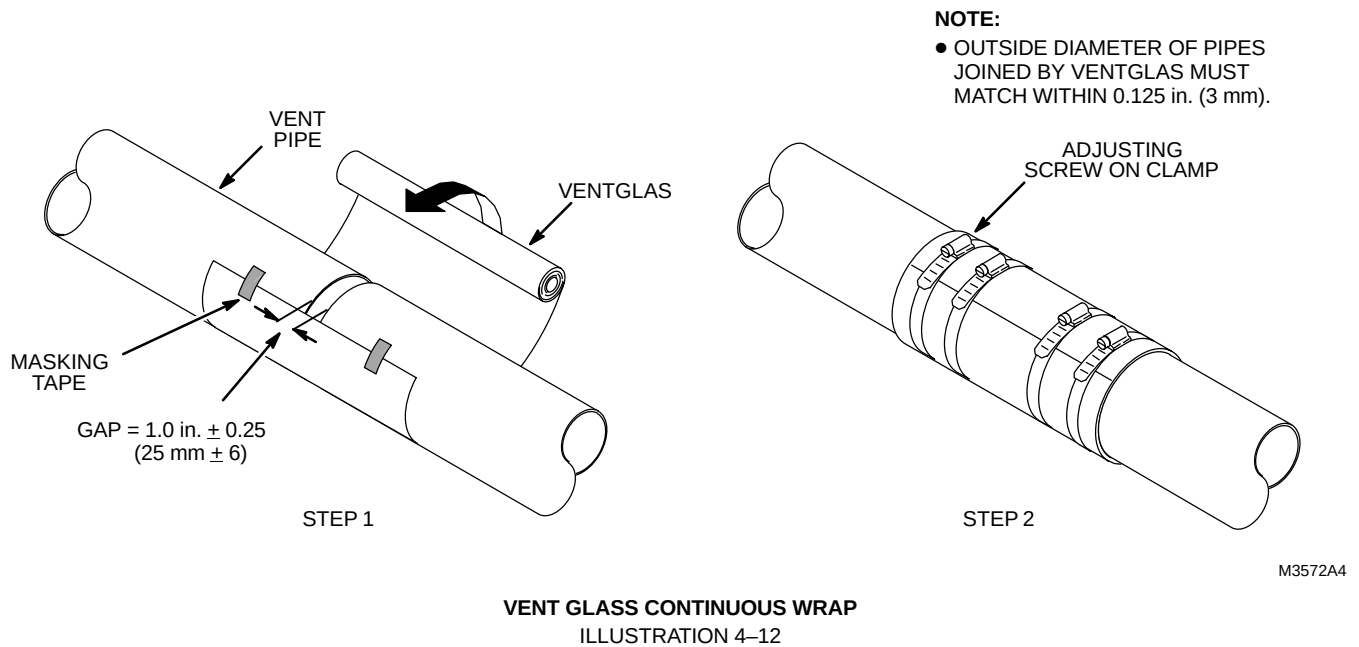
TABLE 4-14
CRYOGENIC VENTING REQUIREMENTS FOR INSIDE MAGNET ROOM

PARAMETER	CRYOGENIC VENTING REQUIREMENTS									
VENT SIZE	- The total pressure drop of the entire cryogenic vent system must be less than 17 psi (117.2 KPa). The calculation starts at the magnet vent interface and ends at the termination point outside of the building. Use Table 4–13 to determine proper vent sizes. - The pressure drop of the RF shield waveguide must be included in the overall calculation. Make sure the RF shielded room supplier provides for 1.5T LCC (Active Shield) magnets 8 inch (203.2 mm) outside diameter straight pipe for the waveguide. Refer to Section 7, RF SHIELDED ROOM, for location of vent pipe/waveguide.									
VENT MATERIAL	- The vent material must be Stainless Steel 304, Aluminium 6061–T6 or Copper type DWV, M or L. - Either tubes or pipes meeting the requirements may be used for venting. The vent pipe must be either seamless or have welded seams. Corrugated pipe must not be used. If necessary, bellows pipe of length less than 1 ft (30 cm) is allowed for thermal expansion joint. - For 1.5T magnets, the vent pipe must be capable of withstanding pressures up to 35 psi (241.4 KPa). - For the waveguide, the vent material must match the outside diameter of the magnet's vent. Refer to Table 4–15. - Venting wall thickness:<table><tr><td>SS 304</td><td>Minimum 0.035 in. (0.89 mm)</td><td>Maximum 0.125 in. (3.18 mm)</td></tr><tr><td>AL 6061–T6</td><td>Minimum 0.083 in. (2.11 mm)</td><td>Maximum 0.125 in. (3.18 mm)</td></tr><tr><td>CU DWV, M or L</td><td>Minimum 0.083 in. (2.11 mm)</td><td>Maximum 0.140 in. (3.56 mm)</td></tr></table>	SS 304	Minimum 0.035 in. (0.89 mm)	Maximum 0.125 in. (3.18 mm)	AL 6061–T6	Minimum 0.083 in. (2.11 mm)	Maximum 0.125 in. (3.18 mm)	CU DWV, M or L	Minimum 0.083 in. (2.11 mm)	Maximum 0.140 in. (3.56 mm)
SS 304	Minimum 0.035 in. (0.89 mm)	Maximum 0.125 in. (3.18 mm)								
AL 6061–T6	Minimum 0.083 in. (2.11 mm)	Maximum 0.125 in. (3.18 mm)								
CU DWV, M or L	Minimum 0.083 in. (2.11 mm)	Maximum 0.140 in. (3.56 mm)								
VENT SUPPORT	- All portions of the cryogenic vent system must be adequately supported. The vent support assemblies must be capable of supporting the weight of the vent system. To ensure the integrity of the RF shield for the magnet room, electrically isolate any support assemblies which are used to support sections of venting between the magnet interface and the isolation joint at the waveguide. - For 1.5T magnets, the vent support assemblies must be capable of supporting the weight of the vent system and 1850 lb. (8229 N) helium flow reaction force at vent elbows. - Non–metallic (i.e. Ventglas) joint must not be used as the support for the vent system.									
VENT CONSTRUCTION	- One dielectric break in the vent system (ie. Ventglas) is required within the Magnet Room. A non–metallic isolation joint must be provided between the GEMS supplied 24 in. (610 mm) long vent pipe and the cryogen vent system ensure the integrity of the RF shield for the Magnet Room. - For vent systems with offsets, a thermal expansion joint may be necessary to allow for the thermal expansion and contraction of the vent system. - All joints except 1 non–metallic isolation (ie. Ventglas) must be welded or brazed. No clamped, sealed flanges permitted. - All isolation/thermal expansion joints except 1 non–metallic isolation (ie. Ventglas) must be able to withstand temperatures from 4.5 K (–451° F or 268° C) and above. For 1.5T magnets, these joints must also withstand pressures up to 35 psi (241.4 KPa). - GE requires Ventglas material to be used for 1 isolation joint, which may serve as a non–metallic thermal expansion joint. Rubber soil pipe couplings are not acceptable for this application. When using Ventglas material to join 2 pipes together make sure the gap between the pipes is 1.0 ± 0.25 inch (25.4 ± 6 mm) and use a continuous wrap technique during installation. See Illustration 4–12. Use 2 hose clamps on each side of the joint for securing the Ventglas to the vent pipe. - To prevent condensation during magnet ramping, the vent must be insulated with 1.5 inch (38 mm) thick flexible unicellular insulation. For appearances, all exposed insulation should be covered with white PVC jackets. Note Access must be available to the non–metallic isolation joint located within 24 in. (610 mm) of the Magnet Vent Adapter for annual inspection and/or maintenance by the customer.									

4-9-2 Requirements For Inside Magnet Room (Continued)

TABLE 4-15
MAGNET TYPE MATRIX

MAGNET TYPES	DIMENSIONS LxWxH inches (mm)	HELIUM VOLUME gallons (liters)	PEAK HELIUM FLOW DURING QUENCH ft ³ per min (m ³ per min)	MAGNET VENT OD inches (mm)	VENT LOCATION ON MAGNET SEE SECTION 7-2-2
1.5T LCC (Active Shield) Magnet	69.7x81.15x91.54 (1770x2061x2325)	520.0 (1970)	2737 (77.50)	8 (203.2) Floating Flange	Illustration 7-1



4-10 ALARM DEVICES, WATER SENSORS AND THERMOSTATS

4-10-1 System Cabinet

For Signa TwinSpeed with EXCITE Technology the System Cabinet has 40°C and 50°C temperature sensors. When the temperature in the System Cabinet reaches 40°C a Warning Message will be displayed on the Operator Workspace. When the temperature in the System Cabinet reaches 50°C or above a Warning Message will be displayed on the Operator Workspace and the system scanning operation will be inhibited. Normal scanning operation will be allowed when the System Cabinet temperature returns to normal. Any external alarm device for room temperature monitoring must be supplied by the customer.

For Signa TwinSpeed the System Cabinet has one temperature sensor and one control module which sounds an alarm located in the System Cabinet when temperature reaches 94°F (34.4°C). After 3 minutes of alarm condition PDU will revert to full off condition. Any external alarm device other than mentioned above must be supplied by the customer.

4-10-2 Water Sensor Alarm and Floor Drain

It is recommended that customer supplied water sensor alarms and floor drain be located on floors where water cooled cabinets are positioned, especially under raised flooring.

4-10-3 Pneumatic Patient Alert

The Pneumatic Patient Alert Control Box provides an audible and visual alarm near the operator when the patient depresses the hand held squeeze bulb. The control box is to be mounted with consideration for ease of use by operator, remaining in sight of operator, and remaining within 5 ft. (1.5 m) of an electrical outlet. Note, an outlet on the Operator Workspace may be used. Options for control box location include mounting box vertically (on a wall or other vertical surface), horizontally (place box on a counter top, desk top, or other horizontal surface), or under a shelf within sight of operator.

4-11 AMBIENT RADIO FREQUENCY INTERFERENCE (RFI)

The MR System utilizes spatially encoded radio frequency information to create the MR image. Therefore, it is sensitive to ambient RFI. To protect the MR from ambient RFI (as well as the local environment from Magnetic Resonance RF), all sites require a 100 dB RF Shield, refer to Section 7, RF SHIELDED ROOM, for exact requirements. It is very unlikely that local signals will affect an MR System with a properly designed and installed RF Shield. During the site evaluation visit, GE notes the location of nearby sources of RFI and will advise if further information or on-site testing is required. Most sites do not require on-site testing. Listed in Table 4-16 are the recommended centerband and bandwidth frequencies to be used when measuring radio frequency interference. This table includes those frequency bands which are important for both proton imaging and spectroscopy.

TABLE 4-16
RADIO FREQUENCY SURVEY SPECIFICATIONS

ISOTOPE	1.5T SYSTEM	
	BANDCENTER MHz	BANDWIDTH Hz
¹ H	63.86	916,138
¹⁹ F	60.12	981,882
³¹ P	25.88	390,296
²³ Na	16.90	242,773
¹³ C	16.06	233,925

When required, RFI site surveys are to be performed by cycling through the preceding frequency bands and a broad band range from 10MHz–100MHz. Special emphasis, however, should be placed on the 1H band since this is used in proton imaging. The RFI site survey should be performed for a length of time necessary to determine, within a reasonable degree of certainty, that the RFI noise at the site will not exceed the 100 db attenuation provided by the RF shielded room. Note that any RFI site survey no matter how thorough, will not preclude the possibility of future or unmeasured RFI caused by new or intermittent sources.

The ambient RF noise measured should be less than 100 millivolt per meter (100 dB microvolt per meter). When a RFI site survey is required, it must be completed before the purchase and installation of the RF shielded room.

To ensure that 100 millivolt (or greater) RF noise peaks outside the bandwidths specified above do not actually extend into these bandwidths and exceed the 100 millivolt limit, adjust the resolution of the test equipment (spectrum analyzer) according to the equation:

$$BW \text{ (resolution)} = f_0 / 50$$

where: BW = Bandwidth (resolution)

f_0 = Center frequency (for 1H: at 1.5 Tesla 63.86 MHz)

4-12 POLLUTION

The site must be clean prior to delivery of the equipment. Although individual components have filters for optimum air filtration, care should be taken to keep air pollution to a minimum.

Since static discharge can cause system failures or affect its operation, carpeting should be of the anti-static type or treated with an anti-static solution.

When cleaning tile floors, do not use steel wool which could enter cabinet enclosures and cause internal shorts.

The computer/equipment area requires that the air be filtered to remove 90 percent of all particles down to 10 microns and 80 percent of all particles from 10 to 5 microns in size.

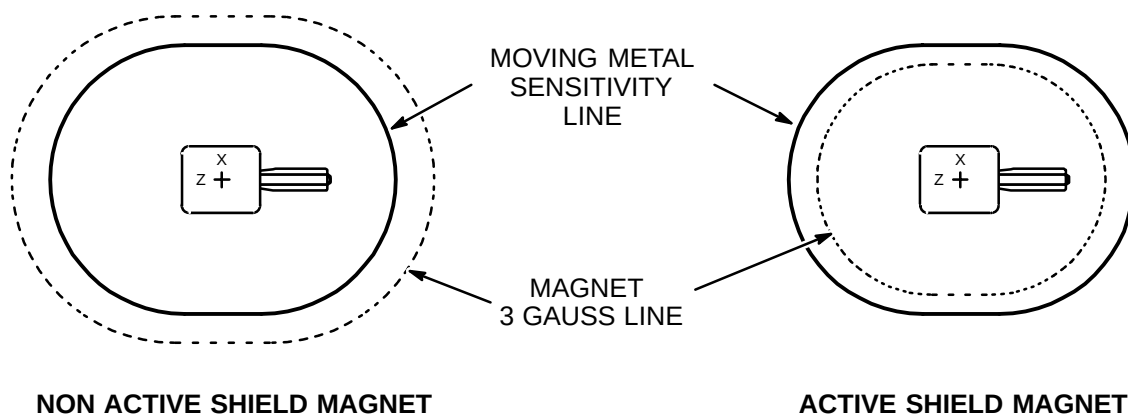
4-13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS

Note

Refer also to Table 2-1, PROXIMITY LIMITS, for additional proximity limitations.

Explanation Of Active Shield Changing Magnetic Environment (Moving Metal Phenomenon)

Prior to the introduction of Active Shield magnets it was necessary to keep moving metal objects outside of the magnet's field. With Active Shield magnets having very small 3 gauss line plots, a new parameter became the limiting factor for moving metal objects proximity to the magnet. In this case, now the object's magnetic moment interacts with the magnet's field. See Illustration 4-13.



MOVING METAL SENSITIVITY LINE COMPARISON
 ILLUSTRATION 4-13

Definition Of Moving Metal

Moving metal means metal objects that move inside of the moving metal sensitivity line during system scans. For example, cars being driven inside the moving metal sensitivity line are moving metal. However, if a car or a dumpster is within the moving metal sensitivity line and **does not** move during scans, then it is not an issue. Note, the 3 gauss line proximity limit for metal objects still applies to Active Shield magnets.

4-13-1 Steel Objects Categories And Requirements

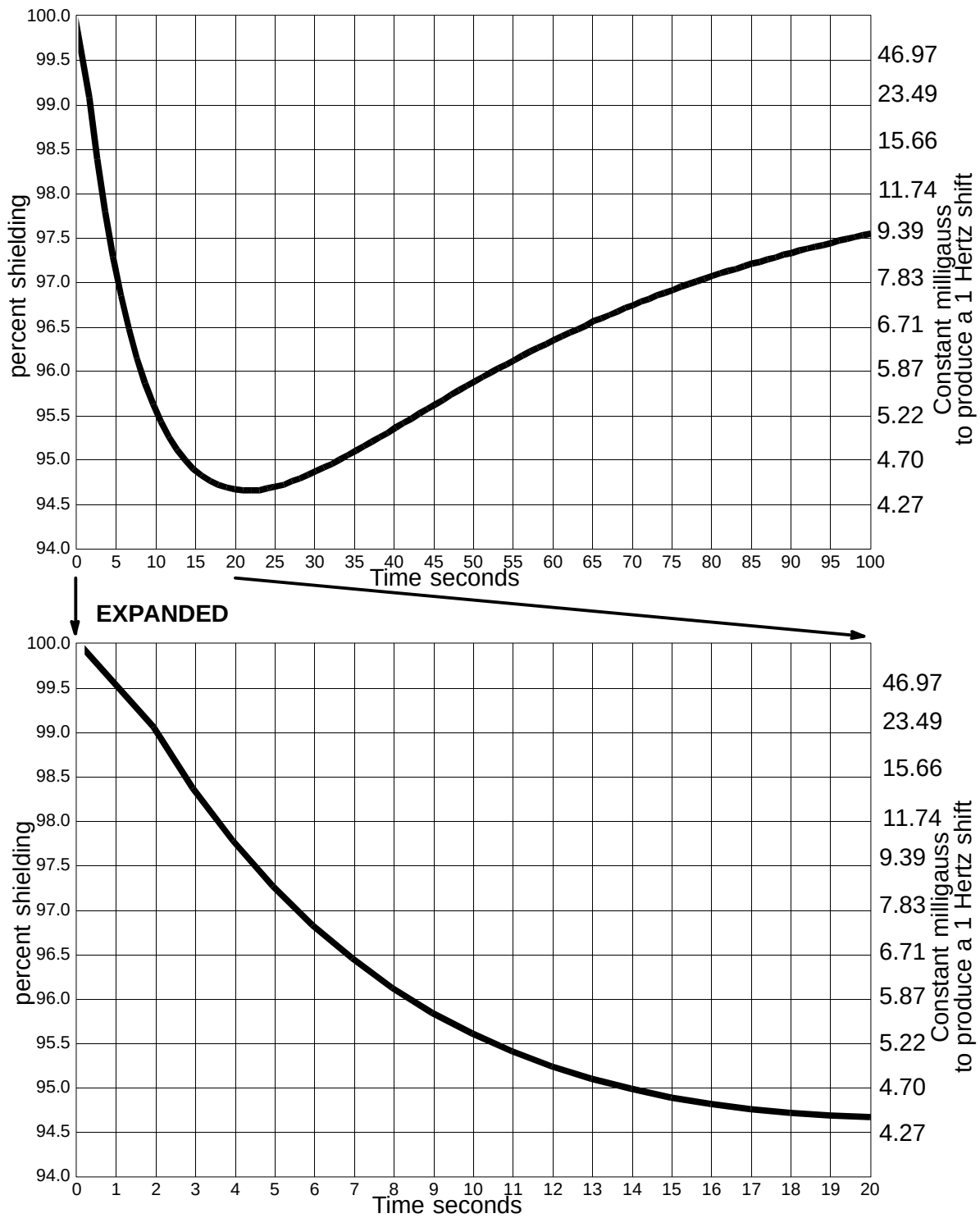
Refer to Table 4-17 for LCC (Active Shield) magnet moving metal requirements and the appropriate illustrations:

- For 1.5T LCC (Active Shield) magnet see Illustrations 4-16 & 4-17.

TABLE 4-17
LCC (ACTIVE SHIELD) MAGNET MOVING METAL REQUIREMENTS

STEEL OBJECTS CATEGORY	DEFINITION OF DISTANCE LOCATION	MINIMUM DISTANCE RADIAL X AXIAL ft (m) See Note 1
Objects 100 – 400 lbs	distance from isocenter radial x axial (See Note 1)	3 Gauss line 7.87 x 13.45 (2.4 x 4.1)
CARS, MINIVANS, VANS, PICKUP TRUCKS, AMBULANCES	distance from isocenter measured to center of driving or parking lane radial x axial (See Note 1)	15.5 x 21 (4.72 x 6.40)
BUS, TRUCKS (UTILITY, DUMP, SEMI)	distance from isocenter measured to center of driving or parking lane radial x axial (See Note 1)	18.1 x 24.5 (5.52 x 7.47)
Objects > 400 lbs, ELEVATORS, TRAINS, SUBWAYS	Place a directional probe (e.g. flux gate sensor) at isocenter of proposed magnet location aligned along the Z-axis. Measure p-p magnetic field change (dc).	See Illustration 4-14 and See Example in Note 2
Note 1 Radial distances are magnet X and Y axis. Axial distances are magnet Z axis. 2 EXAMPLE: For Moving Metal Requirements of objects > 400 lbs category you can use the time history of the occurrence to determine what milligauss level to use. A. If the site has elevators/counter weights near the magnet and the elevator can stop on the floors for longer than 20 seconds (which is usually the case), you need to use a shielding factor of 94.7% and a resulting peak-to-peak milligauss reading of 4.43. B. If the site has a subway nearby and the field disturbance is less than 5 seconds, you can use a shielding factor of 97.3% and a resulting peak-to-peak milligauss reading of 8.39. C. Therefore, be conservative and use 4.43 milligauss peak-to-peak.		

4-13-1 Steel Objects Categories And Requirements (Continued)



ACTUAL AXIAL SHIELDING PERFORMANCE FOR LCC (ACTIVE SHIELD) MAGNET
 ILLUSTRATION 4-14

4-13-2 Distances For AC Power Lines, Transformers And Electric Motors

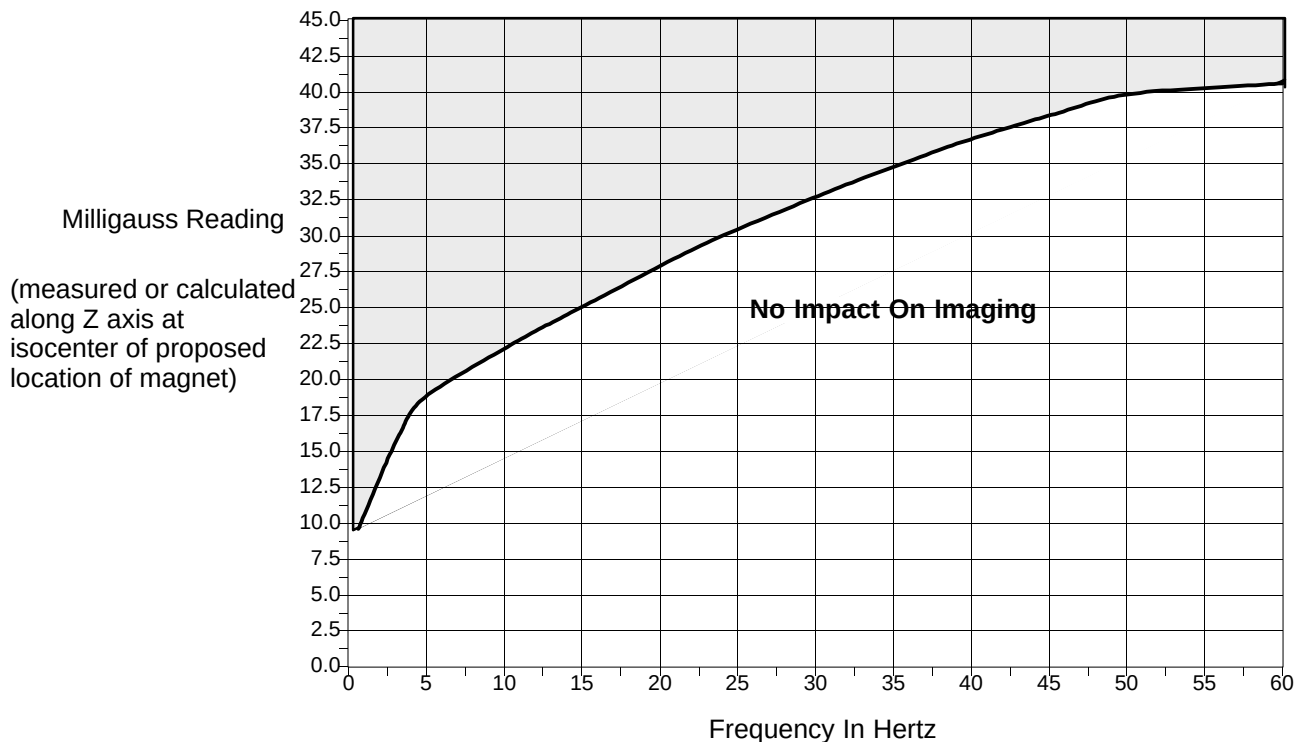
In general most AC equipment in sites is not an issue if it kept outside the 5 gauss line. If a site has large AC equipment (building mains, substations, electric trains, or subways) see Illustration 4-15 and calculate or measure the field along the Z axis at the magnet isocenter.

Electrical currents flowing in high voltage power lines, transformers, and large generators or motors near the magnet can affect the magnetic field homogeneity that is essential to the proper performance of the MR System. Although it is highly unlikely that induced magnetic fields will be a problem, possible sources of AC interference are identified by GE during the site evaluation visit. GE will analyze this information and advise if further shielding or site rearrangement are necessary.

Magnetic field interference at 50 or 60 Hz must not exceed 40 milligauss RMS at the magnet location. The following equation can be used as a general guide in determining allowable current in feeder lines at a given distance from the magnet isocenter.

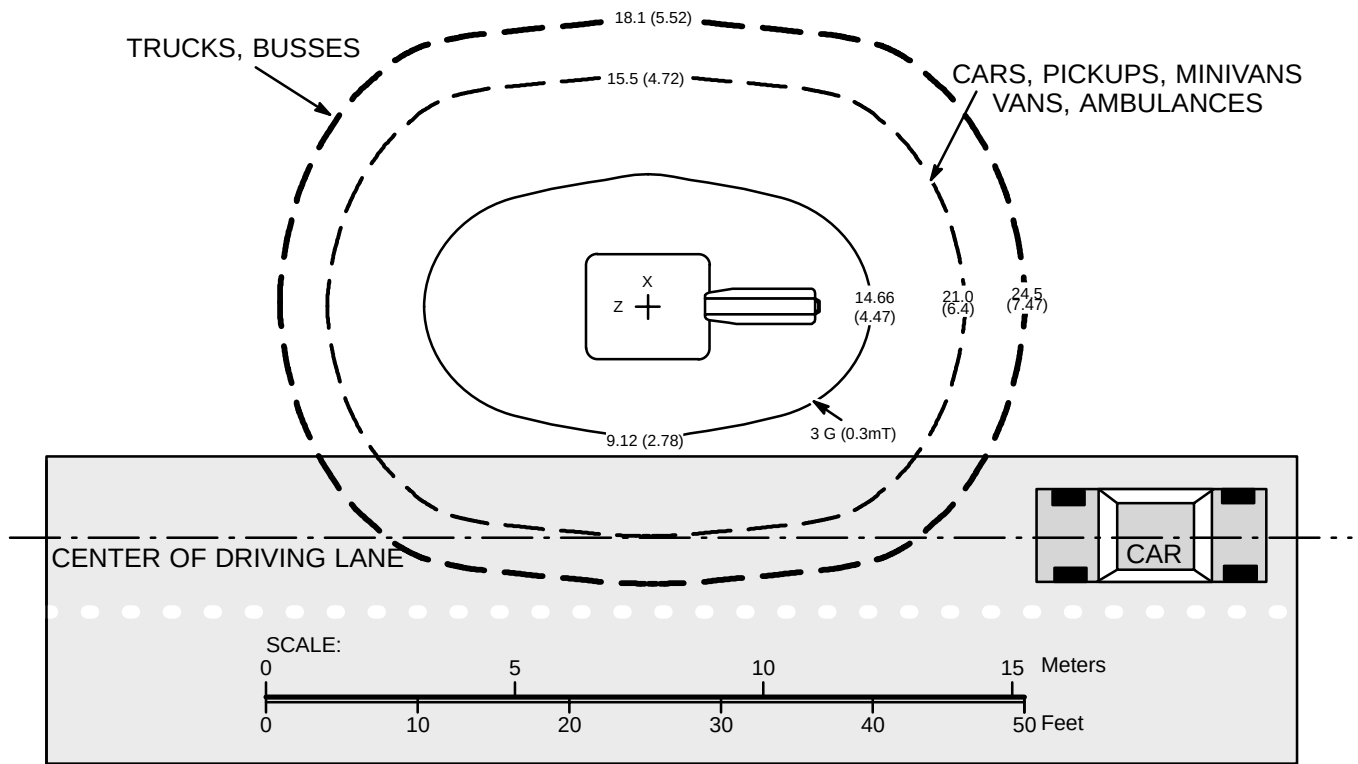
$$I = \frac{20X^2}{S}$$

- where: I = Maximum allowable RMS single phase current (in amps) or maximum allowable RMS line current (in amps) in three phase feeder lines
 S = Separation (in meters) between single phase conductors or greatest separation between three phase conductors
 X = Minimum distance (in meters) from the feeder lines to isocenter of the magnet

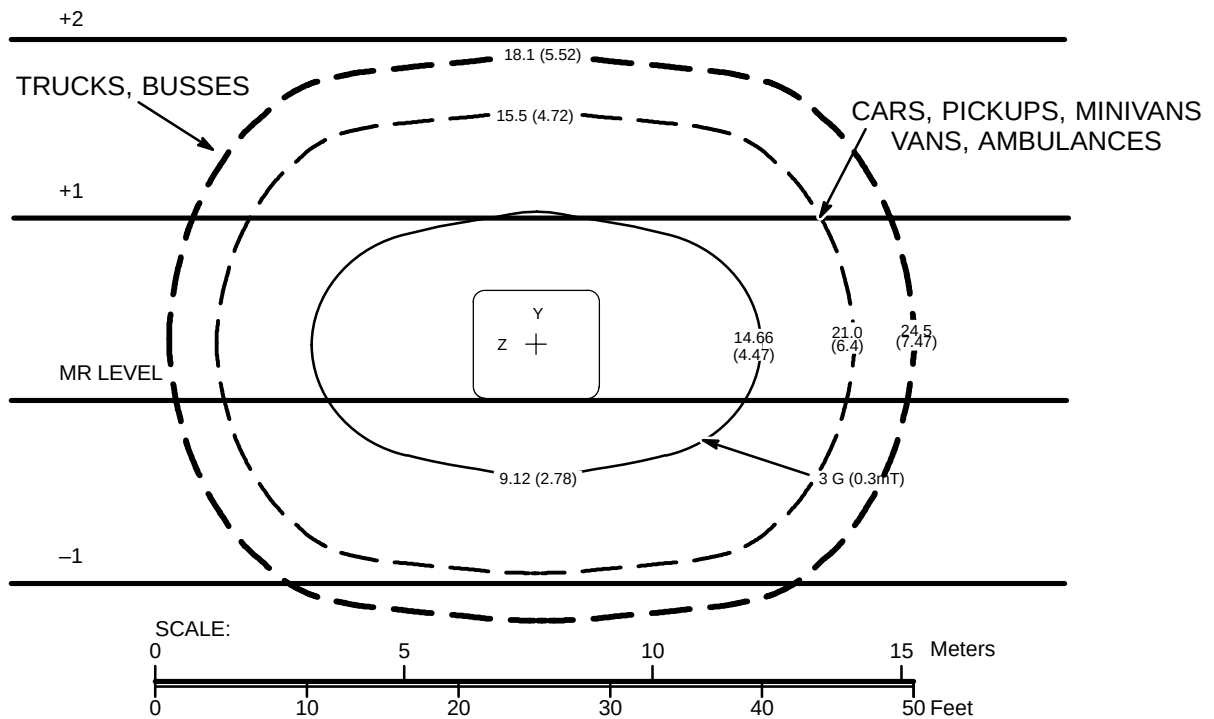


ALLOWABLE MILLIGAUSS VS LINE FREQUENCY FOR AC EQUIPMENT

ILLUSTRATION 4-15



1.5 TESLA LCC (ACTIVE SHIELD) MOVING METAL SENSITIVITY LINE PLOT (MR CENTER LEVEL)
 ILLUSTRATION 4-16



1.5 TESLA LCC (ACTIVE SHIELD) MOVING METAL SENSITIVITY LINE PLOT (VERTICAL VIEW)
 ILLUSTRATION 4-17

4-14 CONSTRUCTION MATERIALS

The following recommendations are for maintaining field homogeneity of the magnet. All construction must comply with local and national building codes.

Note

When welding in an MR room with system equipment installed, the return path for the welding must be in very close proximity to the welding. The close proximity is needed to make sure the welding currents do not cause damage to the system. Never use the building structure as a return path for welding.

4-14-1 Floors

For the 1.5T Active Shield magnet, the magnet room floor should be poured slab on grade with polypropylene fiber impregnated or epoxy reinforced concrete. Non-magnetic stainless steel rebar or fiberglass rebar may also be used as a reinforcing material. Steel reinforcing rods or corrugated iron sheets should be avoided especially within the 50 gauss for the 1.5T Active Shield magnets. If these materials exist at the site, or if installation of these materials is contemplated, they must be taken into account in the structural steel evaluation of the site. Refer to Section 3, MAGNETIC FIELD CONSIDERATIONS, for more information. If necessary, systems with the 1.5T Active Shield magnet can correct for some steel in the floor. This includes steel rebars and other steel building components within a 10 foot by 10 foot (3.1 meter by 3.1 meter) area directly below the magnet. Table 4-18 illustrates the varying limits of mass of steel in close proximity to the magnet isocenter when using normal system shimming techniques. The data is based on a square area located directly beneath the magnet and calculating an equivalent density from the total volume of existing structural steel. In most cases, an I-Beam located directly beneath the magnet is prohibited. For the 1.5T Active Shielded magnets, a single I-Beam larger than W8x40 (beam 8 inches wide and 40 lbs/ft weight) or metric HE200B (200 mm wide wings steel beam and 61 kg/linear meter), should be kept a minimum of 52 inches (1321 mm) from the magnet isocenter. In any case, the values listed Table 4-18 should not be exceeded.

TABLE 4-18
STEEL MASS PROXIMITY TO MAGNET ISOCENTER

MAGNET TYPE	DISTANCE FROM MAGNET ISOCENTER in. (mm)	DISTANCE BELOW TOP SURFACE OF FLOOR in. (mm)	LIMITS OF STEEL MASS lbs/ft ² (kg/m ²)
1.5T LCC (Active Shield) See Note 1	42 (1067)	0 (0)	0 (0)
	45 (1143)	3 (76)	2 (9.8)
	47 (1194)	5 (127)	3 (14.7)
	52 (1321)	10 (254)	8 (39.2)
	55 (1397)	13 (330)	20 (98.0)
Note 1 If any of the steel mass proximity to magnet isocenter limits are exceeded for the LCC Magnet then the 1000 Amp Magnet Ramp Power Supply must be used to ramp the LCC magnet to field. Refer to Section 10-5 Installation Equipment . When the 1000 Amp Magnet Ramp Power Supply is needed, notify the GE Field Service Engineer to arrange for Power Supply delivery.			

Steel rebar must not be positioned in such a manner as to interfere with anchor bolt locations for the magnet or magnet room equipment, refer to **Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** and **Section 7-8 MAGNET ROOM EQUIPMENT MOUNTING**.

4-14-2 Walls, Ceilings, and Fixtures

General

Standard steel nails, screws, and other hardware are acceptable if properly secured. Any loose steel objects can be violently accelerated into the bore of the magnet. Careful thought should be given to the selection of light fixtures, cabinets, wall decoration, etc. to minimize this potential hazard. For safety, all **removable** items within the magnet room such as switch box cover plates, light fixture components, mounting screws, etc. must be non-magnetic. If you have a specific question about material, bring it to the attention of your GE Installation Specialist.

1.5T LCC (Active Shield) Magnets

Non-movable steel such as wall studs or HVAC components will produce negligible effect on the magnet.

4-14-3 Electrical conduits

Electrical conduit within the magnet room may be steel provided it is inside walls and ceilings. Note, conduit for a receptacle must be metallic. Ferromagnetic material inside the magnet room could inadvertently become a projectile.

4-14-4 Plumbing pipes and drains

Pipes and drains within the magnet room may be iron, if desired, without significant effect on the magnet homogeneity. For safety, any removable items such as faucet handles, drain covers, etc. must be non-magnetic material such as PVC, copper, or brass. Any magnetic material inside the magnet room could inadvertently become a projectile. Refer to Section 4-9, CRYOGENIC VENTING, for cryogenic vent materials requirements.

4-15 VIBRATION

4-15-1 Definition

Improvements in MR imaging technology have increased imaging capabilities. Certain MR procedures require a stable environment to achieve high resolution image quality. **Environmental site vibration has the ability to affect the magnet phase stability and ultimately image quality.** The vibration effects on image quality can be minimized early in the site planning of the MR suite.

The magnet may be sensitive to vibration in the frequency range of 0.5 to 45 Hz depending on the amplitude of the vibration. In the physical area where the MR system is located, every precaution must be taken to ensure that the vibration is minimized. In the magnet siting area, the structural stability and behavioral characteristics can be assessed. **The vibration tests outlined in Appendix A, MR SITE VIBRATION TEST GUIDELINES can be used to assess the vibration environment. Sites which currently pass the vibration stability criteria may proceed with the installation. Sites which have marginal vibration stability require source isolation or structural modifications.** Then it is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specified limits. With the vibration consultant present, local GE Field Service and/or Installation Specialist must verify the elimination/reduction of all identified sources do improve the vibration environment. GE can assist in interpreting marginal site test results and predicting the impact on system performance. However it is ultimately the customer/architect/engineer responsibility to design site solution.

To minimize the interference, the magnet should be placed on a solid floor, located as far as possible from the following vibration sources:

- parking lots
- roadways
- subways
- trains
- hallways
- hospital physical plants containing pumps, motors, air handling equipment, air conditioning units
- elevators
- heliports.

Note

Please note that other items not listed could also be potential sources of vibration.

Note

Vibration isolation is highly recommended at floor connection points of the air conditioning unit(s) to be installed for the purpose of cooling the MR Suite.

Note

Isolation of the MR magnet is not a recommended solution for reducing environmental vibration.

Vibration measurements should be made when the proposed site is located near any of the sources listed here. Measurements should be made using a spectrum analyzer capable of performing the test guidelines detailed in Appendix A, MR SITE VIBRATION TEST GUIDELINES.

4-15-2 Types of Vibration Image Quality Issues

Vibration capable of affecting MR image quality can be classified as either steady state or transient vibration. Steady state vibration typically refers to disturbances caused by rotating machinery. Examples of machinery known to have previously generated vibration image quality problems are exhaust fans, air conditioning blower units, compressors, pumps (air and water), etc. Steady state vibration would be measurable during long time intervals. Transient vibrations are typically a function of the building structure or the building foundation. Transient vibration would typically be associated with vehicular traffic, pedestrian motion, patient transport, door slamming, etc. Transient vibration would typically occur during short periods of time. The transient event would typically decay from a high vibration amplitude to lower levels in short periods of time.

Magnet Siting Requirement

Improper mechanical coupling of the magnet to the floor can in some circumstances create unique vibration characteristics. Magnets improperly mechanically coupled to the floor may experience steady state and/or transient vibration system stability issues. When the system stability tests do not meet acceptance criteria the vibration testing defined in Appendix A will help to identify if the magnet is properly mounted to the floor. Refer to Section 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM for magnet mounting requirements.

4-15-3 Specifications For 1.5T LCC (Active Shield) Magnets

Note

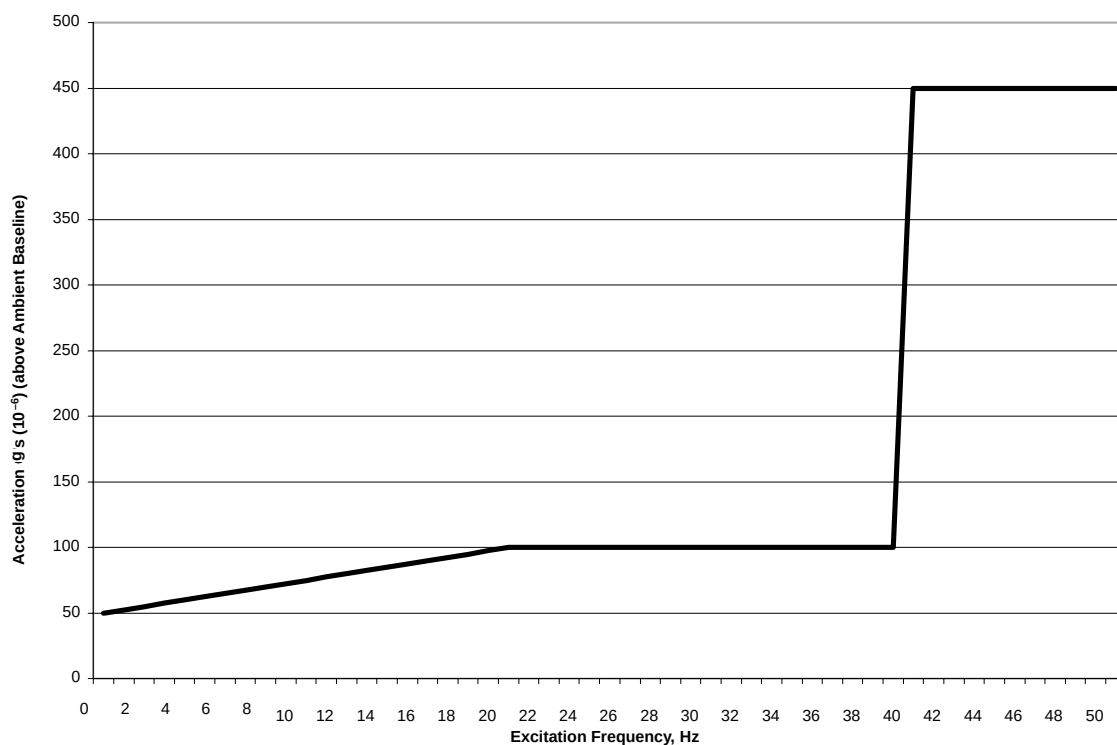
Refer to Appendix A, MR SITE VIBRATION TEST GUIDELINES for definition of ambient baseline.

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline** (refer to Illustration 4-18):

- 5×10^{-5} g rms at 0 Hz ramping to 10×10^{-5} g rms at 20 Hz
- 10×10^{-5} g rms 20-40 Hz
- 45×10^{-5} g rms 40-50 Hz

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.



LCC MAGNET STEADY STATE VIBRATION SPECIFICATIONS
 ILLUSTRATION 4-18

Refer to **Appendix A – MR SITE VIBRATION TEST GUIDELINES** Illustration A-2 for an example applying the steady state specifications with ambient baseline.

4-15-4 Specifications For 1.5T LCC (Active Shield) Magnets

Transient Vibration

Note

Transient vibration analysis requires the elimination of all steady state vibration so as not to mask the transient signal.

Time history transient levels exceeding **500 micro-g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro-g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined in preceding subsection and Illustration 4-18. That is, the vibration consultant must measure and report the transient disturbance of concern. The consultant must determine (assess) the frequency, amplitude, and duration of the transient, then determine whether the disturbance will vibrate the building structure, MR RF Shielded Room, plus the magnet to amplitudes that would exceed the **Steady State Vibration** specifications, refer to preceding subsection and Illustration 4-18.

As an example; a consultant might find transient events with pulse width greater than 5 milliseconds with vibration response (decay) longer than 100 milliseconds which may suggest a need for further analysis. The vibration consultant should document in the report any transient event that may excite the building resonance (including the MR system installed weight).